

Counterexamples to List and Line Decodability Bounds over Prime Fields

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Abstract

Security analyses of hash-based SNARKs use quantitative statements about list decoding and proximity gaps of error-correcting codes. We give counterexamples to several proposed bounds for Reed–Solomon codes over prime fields. The results concern the dependence on the code rate, the gap from capacity, the field size, and the evaluation domain.

For every fixed rational rate $\rho \in (0, 1)$, we construct codes with $2^{\Omega_\rho(n)}$ codewords agreeing with one received word on $(\rho + \eta)n$ coordinates, where $\eta = \Theta_\rho(1/\sqrt{n \log n})$ and $\log_2 p = \Theta_\rho(\sqrt{n \log n})$. The domain is an integer interval embedded in $\mathbb{F}_p$, and the radius lies strictly below the characteristic-based Elias bound. Thus no $\exp(O_\rho(1/\eta))$ list bound holds uniformly over prime fields and evaluation domains. The construction follows by counting subsets with matching moments. Concentration and exact polynomial-weight certificates sharpen the finite bounds.

We also separate list size from the number of nearby points on an affine line. A large list gives an affine line with many nearby points and no correlated agreement. A separate lift preserves a selected-list obstruction at a fixed positive gap and arbitrarily large lengths. Finally, at gap $1/n$, we construct codes whose maximum list size is exactly $n - k - 1$, but with $\binom{n}{k+1}$ uniquely nearby points on one affine line. This parameter count attains an unconditional puncturing bound at that radius. The same generic domains have an exact list-size profile at every integer Hamming radius below $n - k$; at fixed positive rates, this profile follows from known upper and lower bounds.

Finite constructions on the norm-one circle give examples in the polynomial spaces used by Circle STARKs. We apply the general results to the quantitative list and line assumptions in the S-two whitepaper, keeping the coding conclusions separate from claims about a complete protocol’s soundness.

1 Introduction

Many hash-based SNARKs ask the prover to commit to evaluations of low-degree polynomials. The verifier then checks a small number of positions and random linear combinations of the committed words. Analyzing these checks requires quantitative information about the underlying code. How many low-degree polynomials can be close to one word? How many points on an affine line can be close to the code? When do the nearby polynomials imply a common agreement set for the original words?

These are coding-theoretic questions, but the numerical answers enter soundness calculations. A bound that is polynomial in the block length may be adequate for an asymptotic theorem and inadequate for a claimed concrete security level. Likewise, a bound proved for random evaluation domains does not automatically apply to a prescribed subgroup. The field size, domain, and order of quantifiers must be explicit.

This paper gives counterexamples to quantitative list and line bounds over prime fields. The main results are independent of any particular proof system. We first prove the general statements, then give finite circle-code examples and apply them to the S-two analysis.

A useful starting point is the root bound. A nonzero polynomial of degree less than k has fewer than k roots, so two distinct Reed–Solomon codewords cannot agree in k positions. This proves a minimum-distance guarantee. It does not answer the first question above: many polynomials may each agree with an arbitrary received word in more than k positions, using different agreement sets.

Write n for the number of evaluation points and $\rho = k/n$ for the rate. We consider agreement on at least $(\rho + \eta)n$ positions, equivalently relative distance at most $1 - \rho - \eta$. The gap $\eta > 0$ is the excess agreement over the rate. Throughout the paper, we distinguish the largest list at one received word, the number of nearby parameters on an affine line, and correlated agreement across that line. None of these quantities can be substituted for another without a theorem relating them.

1.1 The main result

Our first result concerns a particularly simple evaluation domain: $0, 1, \dots, n-1$ in a prime field.

Theorem 1.1 (Exponential lists on integer intervals, informal). *Fix any rational rate $0 < \rho < 1$. For arbitrarily large n , there is a prime p , a rate- ρ Reed–Solomon code over $\mathbb{F}_p$, and a received word with $2^{\Omega_\rho(n)}$ nearby codewords, where*

$$\eta = \Theta_\rho\left(\frac{1}{\sqrt{n \log n}}\right), \quad \log_2 p = \Theta_\rho(\sqrt{n \log n}).$$

Every listed codeword agrees with the received word on exactly $(\rho + \eta)n$ positions. The associated distance is strictly below the characteristic-based Elias radius.

The field is large, but its logarithm is sublinear in the block length. The list is much larger than any bound of the form $c_1 2^{c_2 H_2(\rho)/\eta}$ with constants depending only on the rate: the logarithm of that proposed bound is $O_\rho(\sqrt{n \log n})$, whereas the logarithm of our list size is $\Omega_\rho(n)$. Here H_2 is binary entropy. For a fixed rate it is simply a positive constant.

The qualification about the field and domain matters. This is a statement about what can be guaranteed for *every* prime-field Reed–Solomon code. It does not give exponential lists at a fixed positive gap, or when the prime is required to be polynomial in n .

Finite and uniform refinements. The moment count also gives explicit certificates. At length 64 and dimension 32, Proposition 2.4 proves a list of at least 5,074,503,250,115 codewords with 34 agreements. Proposition 2.5 gives an analytic refinement: for each fixed number m of canceled coefficients and fixed limiting rate, concentration improves the full-range counting guarantee by a factor of order $n^{m/2}$. The exact checks certify the finite arithmetic; the proofs establish the asymptotic statements.

1.2 Why the construction works

There is a useful way to reverse the usual root-counting argument. Choose a set A of $r = k + s$ evaluation points, and form its root polynomial

$$F_A(X) = \prod_{a \in A} (X - a).$$

Suppose many choices of A give the same coefficients in degrees $r, r-1, \dots, k$. Let W be this common high-degree part. Then $G_A = W - F_A$ has degree less than k , and it agrees with W at every point of A . Thus each such set produces a member of one decoding list. The problem has become a counting question about leading coefficients.

Those coefficients are determined by the first s power sums of A . When $A \subseteq \{0, \dots, n-1\}$, each power sum is a relatively small *integer*. There are $\binom{n}{r}$ possible sets, but far fewer possible vectors of s power sums. Some vector must occur many times. The crucial point is that we count these vectors over the integers *before choosing the prime*. The count therefore does not grow with the field size.

For a small example, take

$$A = \{0, 1, 6, 7\}, \quad B = \{0, 2, 5, 7\}.$$

Both sets have size four and sum fourteen. Their root polynomials therefore begin with $X^4 - 14X^3$. Subtracting either polynomial from $W = X^4 - 14X^3$ leaves a polynomial of degree at most two. Over any prime larger than seven, on the domain $\{0, \dots, 7\}$ each agrees with W at four positions. The general construction repeats this idea with many more sets and more moments. This example illustrates the mechanism; its two codewords are not the quantitative lower bound of Theorem 1.1.

1.3 Lines of received words

Proof systems often test a random linear combination of several words. This leads to a related question. Suppose many words on an affine line $f_z = f_0 + zf_1$ are close to the code. Must the nearby codewords fit together into a single affine line of codewords? Must f_0 and f_1 agree with two codewords on a large common set of coordinates?

We give three constructions, because these questions have different quantifiers.

Many nearby points without correlated agreement. A large list can be converted into a line by dividing by $X - \tau$, where τ lies outside the evaluation domain. This division lowers the witness degree by one. It also makes the direction word $1/(X - \tau)$ unable to agree with a low-degree polynomial on too many points. A collision count ensures that many members of the original list give different parameters z .

An obstruction that persists at a fixed gap. The gap in Theorem 1.1 shrinks with n . To address asymptotic claims at a fixed gap, we start with one fixed large list and replace each evaluation point by B points on which X^B has the same value. Polynomial composition preserves the rate, relative agreement, and list size. Suitable prime fields exist for every B .

A selection argument then shows that the coefficient of n in a line-decodability threshold must account for this list size. If the coefficient is smaller, no additive term that is $o(n)$ along the resulting family can repair it. This second construction *does* have correlated agreement. It rules out a sublinear remainder along this fixed-gap family.

Small lists with many isolated nearby line points. Large lists are not the only obstruction. At every fixed rate $\rho = k/n$, we also construct codes whose maximum list size is exactly $n - k - 1$, together with a line having $\binom{n}{k+1}$ nearby parameters. Each nearby word has a unique nearby codeword, yet no parametrized codeword line contains more than $n - k$ of those witnesses. The gap is $1/n$. At rate one-half these bounds are $n/2 - 1$, $\binom{n}{n/2+1}$, and $n/2$, respectively. The construction combines an elementary line spanned by consecutive powers with a domain chosen using a known list-decoding theorem for random Reed–Solomon codes. Proposition 5.4 shows that its nearby-parameter count is optimal among lines without correlated agreement at this radius.

1.4 Finite examples on circle codes

We also study a different regime: the fixed prime $p = 2^{31} - 1$ and domains on the norm-one circle. These are structured evaluation domains used by Circle STARKs. At the common parameters

$$n = 2^{26}, \quad \rho = \frac{1}{2}, \quad \eta = \frac{33}{1024} - \frac{2}{n},$$

we obtain the following finite lower bounds:

Object constructed	Guaranteed size
One base-field decoding list	$> 2^{40}$
Distinct nearby parameters on an extension-field line	$> 2^{74}$

The line has no correlated agreement at the specified threshold. The constructions give existence with explicit numerical certificates; they do not enumerate these very large lists. Section 6 explains the idea, and Appendix A gives the coefficient calculations.

1.5 Context and organization

Proximity gaps are established tools in the analysis of low-degree tests [2, 11]. In particular, existing theorems give proximity guarantees in specified radius regimes. Our fixed-rate asymptotic examples have $\eta \rightarrow 0$ and eventually lie beyond the Johnson radius $1 - \sqrt{\rho}$, although they remain below the characteristic-based Elias radius. The distinction between these radii is essential when comparing the results with proved soundness bounds for a low-degree test.

The connection between equal subset moments, elementary symmetric functions, and Reed–Solomon decoding is established prior work; Gandikota, Ghazi, and Grigorescu use this connection in decoding hardness results [7]. The results here use quantitative counts on integer intervals to obtain the stated list and line separations. The number-theoretic splitting theorem used in the lift is also classical.

The generic list-profile upper bound is due to Brakensiek, Gopi, and Makam [5]. At every fixed positive rate, Roth’s ordinary list-decoding bound [6] gives the matching lower bound for all sufficiently large lengths. We include an explicit construction that also covers the remaining finite parameters $k \geq 2$, without claiming priority for that construction. The separation here combines these list bounds with a line containing many isolated nearby points.

Krachun, Kazanin, and Haböck [10] construct large lists and affine-line obstructions on multiplicative subgroups of prime fields, with a gap of order $1/\log n$. Our interval construction addresses a different family of domains and gives lists larger than every fixed exponential in the reciprocal gap. The multiplicative structure of their examples remains a separate advantage. Our finite circle examples concern the evaluation spaces of Circle STARKs [12].

The Proximity Prize’s orbit-pencil upper certificate also uses multiplicative fibers and a fixed core [13]. Our circle construction expresses the analogous core through a common anchor and division at two circle points. Appendix A.6 makes this change of presentation explicit. The contest’s field, code, and score are different from the circle specialization considered here.

The distinction between fixed and shrinking gap separates our results from the fixed-gap polynomial-in- n list bound stated in the recent preprint of Jeronimo [8]. Our exponential-list family has a shrinking gap. Our fixed-gap family preserves a fixed finite list of witnesses as n grows; it does not bound the full decoding list by that constant. Neither construction contradicts such a polynomial bound.

Sections 2–5 prove the general list and line results. Section 6 gives finite circle examples. Section 7 explains what these results imply for hash-based SNARK analyses, and Section 8 applies them to S-two. Section 9 states the remaining coding-theoretic questions. The appendices contain the circle proofs, quotient calculations, and finite certificates.

2 Large lists from subsets with equal moments

The construction starts with a simple observation: if two polynomials differ by a polynomial whose roots are a set A , then they agree at every point of A . We will choose many sets A whose vanishing polynomials have the same leading coefficients. Subtracting those common coefficients will give many low-degree polynomials close to one received word.

Notation. For a field F and a set $D \subset F$ of n distinct points, write

$$\text{RS}(F, D, k) = \{(P(x))_{x \in D} : \deg P < k\}.$$

This code has dimension k and rate $\rho = k/n$, provided $k \leq n$. The relative Hamming distance between two words is the fraction of coordinates on which they differ. A list at radius θ consists of all codewords within that distance of a received word. Write $\ell_C(\theta)$ for the largest such list over all received words. We parameterize the radius by

$$\theta = 1 - \rho - \eta.$$

Thus a nearby codeword agrees with the received word on at least a $\rho + \eta$ fraction of the coordinates. We call $\eta > 0$ the gap from $1 - \rho$.

All logarithms are base two, and $H_2(t) = -t \log_2 t - (1 - t) \log_2(1 - t)$ is binary entropy. For characteristic p , define

$$H_p(t) = \frac{H_2(t) + t \log_2(p - 1)}{\log_2 p}.$$

The characteristic-based Elias radius is the solution of $H_p(r_E) = 1 - \rho$ on the increasing branch $[0, 1 - 1/p]$. We will keep our examples strictly below this radius. The simple estimate

$$H_p(\theta) \leq \theta + \frac{1}{\log_2 p} \tag{1}$$

shows that it suffices to have $\eta \log_2 p > 1$. In our prime-field examples $p > n$ and $\rho \geq 1/n$, so $0 \leq \theta < 1 - 1/p$ as required.

2.1 Why matching moments produces a list

Consider the 34-element subsets A of $\{0, \dots, 63\}$. Their vanishing polynomials begin

$$F_A(X) = \prod_{a \in A} (X - a) = X^{34} - \left(\sum_{a \in A} a \right) X^{33} + \left(\sum_{a < b \in A} ab \right) X^{32} + \dots$$

Subsets with the same sum and the same sum of squares have the same three leading coefficients, because

$$\sum_{a < b \in A} ab = \frac{(\sum_{a \in A} a)^2 - \sum_{a \in A} a^2}{2}.$$

Let W be those common leading terms. Each polynomial $W - F_A$ has degree less than 32 and agrees with W at exactly the 34 points of A . Consequently, a large class of subsets sharing these two moments gives a large list in a rate-one-half Reed–Solomon code.

For the general construction we fix the first s moments. It is convenient to use *binomial moments*, $\sum_{a \in A} \binom{a}{j}$, because their integer ranges can be counted exactly. They determine the usual power sums and hence the leading coefficients in exactly the same way.

Lemma 2.1 (A finite list bound). *Let $n \geq 2$, $k \geq 1$, $s \geq 1$, and $t = k + s < n$. Let $p > n$ be prime and $D = \{0, \dots, n-1\} \subset \mathbb{F}_p$. Put*

$$R_j = \binom{n}{j+1} - \binom{n-t}{j+1} - \binom{t}{j+1} + 1 \quad (1 \leq j \leq s), \quad L = \left\lceil \frac{\binom{n}{t}}{\prod_{j=1}^s R_j} \right\rceil.$$

There is a received word with at least L distinct codewords of $\text{RS}(\mathbb{F}_p, D, k)$ at distance exactly $1 - t/n$. If $p > \max_j(R_j - 1)$, its entire list at radius $1 - t/n$ corresponds exactly to one class of subsets with equal integer binomial moments.

Proof. For each t -subset $A \subset D$, record $T_j(A) = \sum_{a \in A} \binom{a}{j}$ for $1 \leq j \leq s$. The minimum and maximum values of T_j occur on the first and last t integers in D . The identity $\sum_{a=0}^{b-1} \binom{a}{j} = \binom{b}{j+1}$ gives their values as

$$\binom{t}{j+1} \quad \text{and} \quad \binom{n}{j+1} - \binom{n-t}{j+1}.$$

There are therefore at most $\prod_j R_j$ moment vectors. By pigeonhole, at least L subsets share one vector.

Equal binomial moments imply equal power sums through degree s : each X^j is an integer linear combination of $\binom{X}{0}, \dots, \binom{X}{j}$, and all subsets have size t . Newton identities then imply equal elementary symmetric functions through degree s . These identities are valid modulo p since $p > n > s$. Thus the monic polynomials $F_A(X) = \prod_{a \in A} (X - a)$ have the same coefficients in degrees $t, t-1, \dots, t-s = k$. Let W be their common degree- $\geq k$ part. Then $P_A = W - F_A$ has degree less than k and agrees with W exactly on A : outside A , each factor of $F_A(x) = \prod_{a \in A} (x - a)$ is nonzero in the field because $p > n$ makes the evaluation points distinct. Different subsets give different codewords, since these polynomials have degree less than n .

For the last assertion, any codeword P in the list makes $W - P$ a monic degree- t polynomial with at least t roots in D . It must therefore be F_A for a t -subset A . Its first s binomial moments are congruent to those of the selected class modulo p . Each lies in an integer interval of width $R_j - 1 < p$, so the congruences are equalities. $\square$

2.2 An exponential list and a subexponential proposed bound

There are exponentially many subsets of a fixed density. Fixing a modest number of moments divides them into far fewer classes. The following choice of parameters makes this discrepancy precise.

Theorem 2.2 (No uniform $\exp(O_\rho(1/\eta))$ list bound). *Fix a rational rate $\rho \in (0, 1)$. For arbitrarily large n , there are prime-field Reed–Solomon codes of exact rate ρ , on $D = \{0, \dots, n-1\}$, with*

$$\eta = \Theta_\rho\left(\frac{1}{\sqrt{n \log n}}\right), \quad \log_2 p = \Theta_\rho(\sqrt{n \log n}), \quad \log_2 \ell_C(1 - \rho - \eta) = \Omega_\rho(n).$$

The radius lies strictly below the characteristic-based Elias radius. Consequently no finite constants $c_1, c_2 > 0$ make

$$\ell_C(1 - \rho - \eta) \leq c_1 2^{c_2 H_2(\rho)/\eta} \quad (2)$$

hold uniformly over prime fields, evaluation domains, and these radii. The same sequence works for every fixed pair c_1, c_2 .

Proof. Let n grow through multiples of the denominator of ρ and set $k = \rho n$. Choose a constant $c > 0$ with $c^2 < 4H_2(\rho)$, and put

$$s = \left\lfloor c \sqrt{\frac{n}{\log_2 n}} \right\rfloor, \quad t = k + s, \quad \eta = s/n.$$

For large n , the hypotheses of Lemma 2.1 hold. Retaining the factorials in the binomial coefficients gives $R_j \leq 2^{\binom{n}{j+1}} \leq 2^{(3n/(j+1))^{j+1}}$. Hence

$$\log_2 \prod_{j=1}^s R_j \leq s + \sum_{j=1}^s (j+1) \log_2 \frac{3n}{j+1} = \frac{s^2}{2} \log_2 \frac{n}{s} + O(s^2 + s \log n) = \left(\frac{c^2}{4} + o(1) \right) n.$$

The middle equality follows by summing $j \log j$, or by integral comparison. The final equality uses $\log_2(n/s) = \frac{1}{2} \log_2 n + O(\log \log n)$. On the other hand, $\log_2 \binom{n}{t} = H_2(\rho)n + o(n)$. The lemma gives

$$\log_2 L \geq \left(H_2(\rho) - \frac{c^2}{4} - o(1) \right) n = \Omega_\rho(n). \quad (3)$$

Choose any constant $A > H_2(\rho)/c$. Bertrand's postulate supplies a prime with $\log_2 p = A\sqrt{n \log_2 n} + O(1)$. Eventually $p > n$ and $\eta \log_2 p \rightarrow cA > H_2(\rho)$. Since $H_2(1 - \rho - \eta) \rightarrow H_2(\rho)$, the sharper entropy estimate $H_p(\theta) \leq \theta + H_2(\theta)/\log_2 p$ proves the Elias condition. In contrast, the logarithm of the right-hand side of (2) is

$$\log_2 c_1 + c_2 H_2(\rho) \frac{n}{s} = O_{\rho, c_1, c_2}(\sqrt{n \log n}) = o(n).$$

It is eventually smaller than the logarithm of the constructed list. $\square$

The rate is fixed, while the gap decreases and the prime grows. The theorem leaves open stronger bounds under restrictions on the domains or on the relation between field size and block length.

Finite examples. At $p = 2^{127} - 1$, $n = 64$, $k = 32$, $s = 2$, Lemma 2.1 gives

$$L > 2^{35}.$$

Proposition 2.3 below strengthens this to $L \geq 5074503250115 > 2^{42}$ by Proposition 2.4. Here $\eta = 1/32$, so the choice $c_1 = c_2 = 1$ in (2) gives only 2^{32} ; the improved ratio exceeds 1181. The exact inequality $p^2 > 2^{64}$ verifies the sufficient Elias condition. At $p = 2^{521} - 1$, $n = 4096$, $k = 2048$, $s = 8$, the same formula gives $L > 2^{3600}$, compared with 2^{512} . The exact inequality $p^8 > 2^{4096}$ verifies Elias. At $c_1 = 1$, this example forces $c_2 > 7$. A larger example over the same prime forces $c_2 > 22$. Appendix D.1 records the exact parameters and counts. As a small check of the algebra, among the 12,870 eight-subsets of $\{0, \dots, 15\}$, the binomial-moment class $(56, 252)$ has exactly 22 members. Over $p = 2^{31} - 1$, they give 22 distinct degree- < 6 polynomials, each agreeing with the common word at exactly eight positions.

2.3 Sharper finite bounds from conditional moments

The product of moment ranges in Lemma 2.1 counts many combinations without using how the moments depend on each other. For two moments, one can first fix the sum of the selected integers, then bound the spread of the second binomial moment within that class. Both steps can be computed exactly without enumerating the subsets.

Proposition 2.3 (Conditional range and dispersion bounds). *Let $3 \leq t < n$, $k = t - 2$, and $p > n$ be prime. For an integer q , let $\mathcal{A}_q$ be the family of t -subsets of $\{0, \dots, n - 1\}$ with sum q , and put*

$$C_q = |\mathcal{A}_q|, \quad Y_A = \sum_{a \in A} \binom{a}{2}.$$

For $C_q > 0$, let l_q and u_q be the minimum and maximum of Y_A in this family. There is a received word for $\text{RS}(\mathbb{F}_p, \{0, \dots, n - 1\}, k)$ whose list at radius $1 - t/n$ has size at least

$$\left\lceil \frac{C_q}{u_q - l_q + 1} \right\rceil. \quad (4)$$

More generally, choose an integer $c \in [l_q, u_q]$ and an integer $r \geq 0$. Define

$$D_q(c) = \sum_{A \in \mathcal{A}_q} (Y_A - c)^2, \quad W_q(c, r) = \min(u_q, c + r) - \max(l_q, c - r) + 1.$$

Then a possibly different received word has a list of size at least

$$\left\lceil \frac{\max\{0, C_q - \lfloor D_q(c)/(r + 1)^2 \rfloor\}}{W_q(c, r)} \right\rceil. \quad (5)$$

Every codeword produced by these bounds agrees in exactly t positions.

Proof. Pigeonhole over the integer values of Y_A proves (4). For the second bound, every subset with $|Y_A - c| > r$ contributes at least $(r + 1)^2$ to $D_q(c)$. Thus at most $\lfloor D_q(c)/(r + 1)^2 \rfloor$ subsets lie outside the specified interval. The remaining subsets occupy at most $W_q(c, r)$ integer values, proving (5) by pigeonhole. In either case, a class shares its first two binomial moments. The polynomial construction in Lemma 2.1 applies. $\square$

Exact computation. Process the integers $0, \dots, n - 1$ in order. For each subset size and first-moment sum, store its count, the minimum and maximum of Y , and the two totals $S = \sum Y$ and $Q = \sum Y^2$. On adding the integer j , put $b = \binom{j}{2}$. A predecessor record with count C contributes

$$C, \quad S + bC, \quad Q + 2bS + b^2C$$

to the new count and totals; its minimum and maximum each increase by b . Merge the contributions from taking and omitting j . The desired squared deviation is then $D_q(c) = Q - 2cS + c^2C$. There are $O(nt^2)$ possible size–sum states and n stages, giving $O(n^2t^2)$ integer operations and $O(nt^2)$ records. All stored integers have $O(n + \log n)$ bits. No assumption that the second moment fills its entire range is needed.

A stronger length-64 certificate. For $n = 64$ and $t = 34$, conditioning on $q = 1061$ gives $C_q = 8558724088132137$, $l_q = 17784$, and $u_q = 25714$. The range bound alone gives 1079148163931 codewords. The dispersion bound is stronger. For $q = 1071$, take $c = 22138$ and $r = 1076$. The exact counts are

$$\begin{aligned} C_q &= 8634334451040106, \\ D_q(c) &= 3337241714848691877182, \\ C_q - \left\lfloor \frac{D_q(c)}{1077^2} \right\rfloor &= 5757225839350346, \quad W_q(c, r) = 2153. \end{aligned}$$

Consequently,

$$\ell_C(15/32) \geq \left\lceil \frac{5757225839350346}{2153} \right\rceil = 2674048230075 > 2^{41}, \quad C = \text{RS}(\mathbb{F}_p, \{0, \dots, 63\}, 32). \quad (6)$$

This holds for every prime $p > 64$. At $p = 2^{127} - 1$, the original certificate $p^2 > 2^{64}$ keeps the radius strictly below Elias. The dispersion bound exceeds the product-range bound by a factor of 53, and exceeds $2^{1/\eta} = 2^{32}$ by a factor of 622. These are certified list lower bounds; neither is an exact maximum list size or an enumeration of the large moment class.

The script `checks/verify_conditional_moments.py` checks all counts, extrema, and moment totals against exhaustive enumeration for 66 smaller parameter pairs, covering 32,180 subsets. It independently verifies both displayed conditional counts using the coefficient identity

$$[y^t] \prod_{j=0}^{n-1} (1 + yx^j) = x^{\binom{t}{2}} \prod_{j=1}^t \frac{1 - x^{n-t+j}}{1 - x^j}.$$

Summing the conditional moment totals also agrees with the direct one-element and two-element inclusion counts over all t -subsets.

Signed polynomial weights. The interval argument can be sharpened further: instead of charging every retained moment value equally, assign a polynomial weight to each value. Negative weights outside favored ranges are allowed and improve the bound.

Proposition 2.4 (Polynomial-weight refinement). *Use the notation of Proposition 2.3, fix an integer c , and put*

$$M_j = \sum_{A \in \mathcal{A}_q} (Y_A - c)^j.$$

For any polynomial $P(Z) = \sum_{j=0}^d b_j Z^j$ with rational coefficients, define

$$S_P = \sum_{y=l_q}^{u_q} \max\{P(y - c), 0\}.$$

If $S_P > 0$, there is a received word whose list at radius $1 - t/n$ has size at least

$$\max \left\{ 0, \left\lceil \frac{\sum_{j=0}^d b_j M_j}{S_P} \right\rceil \right\}. \quad (7)$$

Each resulting codeword agrees in exactly t positions.

Proof. Let N_y count the members of $\mathcal{A}_q$ with second moment y , and let $L = \max_y N_y$. Since $0 \leq N_y \leq L$,

$$\sum_j b_j M_j = \sum_y N_y P(y - c) \leq L \sum_y \max\{P(y - c), 0\} = L S_P.$$

The equal-moment construction then gives the asserted codeword list. No positivity assumption on P over the entire moment range is needed. $\square$

Taking $P(Z) = h^2 - Z^2$ gives the particularly simple certificate

$$L \geq \left\lceil \frac{C_q h^2 - D_q(c)}{h(4h^2 - 1)/3} \right\rceil \quad \text{when } C_q h^2 > D_q(c).$$

Here the denominator sums positive weights over all integers, so it remains a valid upper bound if the actual moment range is shorter. The preceding $q = 1071$, $c = 22138$ data with $h = 1077$ already give $L \geq 4009211058641$ without computing any further moments.

An exact length-64 certificate. Continue with $n = 64$, $t = 34$, $q = 1071$, and $c = 22138$. The exact second-moment range is $[17969, 26129]$. Define the monic degree-20 polynomial $P(Z) = \prod_{r \in R} (Z - r)$, where

$$R = \{-4169, -2688, -2687, -2062, -2059, -1507, -1472, \\ -1038, -868, -643, 647, 870, 1039, 1464, 1498, \\ 2035, 2038, 2637, 2638, 3991\}.$$

Exact evaluation of (7) yields

$$\ell_C(15/32) \geq 5074503250115 > 2^{42}, \quad C = \text{RS}(\mathbb{F}_p, \{0, \dots, 63\}, 32), \quad p > 64. \quad (8)$$

This exceeds the original product-range certificate by a factor of 101, the preceding dispersion certificate by a factor of 1.89, and 2^{32} by a factor of 1181. At $p = 2^{127} - 1$, the same below-Elias certificate applies. These remain list lower bounds, not exact maximum list sizes.

For reproducibility, extend the count recurrence to raw moment totals $T_r = \sum Y_A^r$ for $0 \leq r \leq 20$. Adding an integer a with $b = \binom{a}{2}$ contributes

$$T'_r = \sum_{j=0}^r \binom{r}{j} b^{r-j} T_j, \quad M_r = \sum_{j=0}^r \binom{r}{j} (-c)^{r-j} T_j.$$

The positive-weight denominator is an exact sum over 8161 integers. The files in `moment_certificates/` record all raw and centered moments, polynomial coefficients, numerator, and denominator. The polynomial was discovered numerically, then replaced by the displayed integer-root polynomial. Its proof and certificate verification use only integer arithmetic; numerical optimizer tolerances play no role. The checker compares every moment through degree 20 with exhaustive small-subset enumeration and independently checks the length-64 moments using complementary 30-subsets.

The weighted inequality applies to any finite signature map, but its application to a prescribed evaluation domain requires statistics for that domain's coefficient signatures. The interval moments above do not by themselves supply such statistics for a multiplicative NTT subgroup.

2.4 A uniform concentration bound for several moments

The improvement is not limited to a computed length-64 example. Concentrating the moment signatures near their means saves a polynomial factor for every fixed number of canceled coefficients.

Proposition 2.5 (Concentration of triangular moment signatures). *Let $1 \leq m < t < n$, $k = t - m$, and let $p > n$ be prime. Choose rational coefficients c_{ji} and functions*

$$b_j(a) = \binom{a}{j} + \sum_{i=1}^{j-1} c_{ji} \binom{a}{i}, \quad 1 \leq j \leq m, \quad 0 \leq a < n.$$

For a uniformly chosen t -subset A of $\{0, \dots, n-1\}$, write $B_j(A) = \sum_{a \in A} b_j(a)$ and define

$$\bar{b}_j = \frac{1}{n} \sum_{a=0}^{n-1} b_j(a), \quad V_j = \frac{t(n-t)}{n-1} \left(\frac{1}{n} \sum_{a=0}^{n-1} b_j(a)^2 - \bar{b}_j^2 \right), \quad h_j = \left\lceil \sqrt{(m+2)V_j} \right\rceil.$$

Then $C = \text{RS}(\mathbb{F}_p, \{0, \dots, n-1\}, k)$ satisfies

$$\ell_C(1 - t/n) \geq \left\lceil \frac{2 \binom{n}{t}}{(m+2) \prod_{j=1}^m (2h_j + 1)} \right\rceil. \quad (9)$$

The constructed codewords agree in exactly t positions.

Proof. The inclusion probabilities t/n and $t(t-1)/(n(n-1))$ give $\mathbb{E}B_j = t\bar{b}_j$ and $\text{Var}(B_j) = V_j$. Here $V_j > 0$: b_j is a nonconstant polynomial of degree $j < n$ on n distinct integers, and $0 < t < n$. Chebyshev's inequality and a union bound show that at least a $2/(m+2)$ fraction of all subsets satisfy $|B_j - t\bar{b}_j| \leq h_j$ for every j .

For the first coordinate there are at most $2h_1 + 1$ possible integer values. Given the earlier coordinates, triangular inversion fixes the earlier binomial moments, and B_j differs from the integer $\sum_{a \in A} \binom{a}{j}$ by a fixed rational number. Hence its interval contains at most $2h_j + 1$ possible values. There are at most $\prod_j (2h_j + 1)$ retained signatures. Pigeonhole and triangular inversion give a class sharing its first m integer binomial moments. Lemma 2.1 applies. Rational changes of coordinates are used only for counting; no reduction of their denominators modulo p is required. $\square$

For fixed m and $t/n \rightarrow \rho \in (0, 1)$, choosing all $c_{ji} = 0$ gives $V_j = \Theta_{\rho,m}(n^{2j+1})$ and therefore

$$\ell_C(1 - t/n) = \Omega_{\rho,m} \left(\frac{\binom{n}{t}}{n^{m(m+2)/2}} \right).$$

The product of complete moment ranges instead has order $n^{m(m+3)/2}$ in its denominator. Thus concentration improves that guarantee by a factor of order $n^{m/2}$. This comparison fixes m ; it does not replace the separate estimates when m grows with n .

Two moments with explicit constants. For $m = 2$, take $b_1(a) = a$ and $b_2(a) = \binom{a}{2} - (n-2)a/2$. These centered population functions are uncorrelated, and their subset-sum variances are

$$V_1 = \frac{t(n-t)(n+1)}{12}, \quad V_2 = \frac{t(n-t)(n+1)(n^2-4)}{720}.$$

Consequently (9) implies

$$\ell_C(1 - t/n) \geq \left(\frac{3\sqrt{60}}{8\rho(1-\rho)} + o(1) \right) \frac{\binom{n}{t}}{n^4} \quad \text{if } t/n \rightarrow \rho \in (0, 1).$$

At $n = 64, t = 34$, the exact variances are 5525 and 376805, and $h_1 = 149, h_2 = 1228$. This gives 1102772374154 codewords without the conditional dynamic program. The degree-20 certificate is stronger at that length. The script `moment_certificates/verify_concentration.py` checks the variance identities, triangular signature counts, and finite bounds against exhaustive small-subset enumeration.

3 From large lists to nearby codewords on a line

List-decoding asks about one received word. Line-decodability asks how nearby codewords can vary along an affine family of received words $f(z) = f_0 + zf_1$. Its selected-witness formulation asks whether many chosen nearby codewords must themselves follow one affine rule in z .

Definition 3.1 (Selected-witness line-decodability). Let $C \subseteq F^n$ be linear. Say that C is line-decodable at radius θ , threshold a , and concurrency b if the following holds. For every affine word line $f(z) = f_0 + zf_1$, every set $S \subset F$ with $|S| > a$, and every choice of codewords $p_z \in C$ within distance θ of $f(z)$, some $P_0, P_1 \in C$ satisfy $p_z = P_0 + zP_1$ for more than b parameters $z \in S$.

The word “selected” matters: the property must hold for every choice of the nearby witnesses. Also, equality uses the *same* parameter z on both sides. We call $\{(z, P_0 + zP_1) : z \in F\}$ a parametrized codeword line. Separately, (f_0, f_1) has *correlated agreement* on t coordinates if two codewords agree with f_0, f_1 simultaneously on one set of t coordinates.

We first turn a large list into a family that has many nearby codewords but no such correlated agreement. Division by $X - \tau$ lowers each polynomial’s degree by one. The value at the chosen pole τ determines which parameter on the word line receives that polynomial.

Lemma 3.2 (A rational word line). *Let $D \subset \mathbb{F}_p$ have size n , and let $1 \leq d < t < n$. Suppose N distinct polynomials R_i of degree at most d agree with a common word w on at least t coordinates. Write $Q = p - n > 0$. There is a pole $\tau \in \mathbb{F}_p \setminus D$ for which*

$$f_0(x) = \frac{w(x)}{x - \tau}, \quad f_1(x) = \frac{1}{x - \tau}$$

has a set S of distinct parameters, with a selected nearby codeword $P_z \in \text{RS}(\mathbb{F}_p, D, d)$ at each parameter, such that

$$|S| \geq \frac{NQ}{Q + d(N - 1)}. \quad (10)$$

Each selected codeword has at least t agreements. No pair of codewords agrees jointly with (f_0, f_1) on more than d coordinates, and every parametrized codeword line contains at most $\lfloor (n - d)/(t - d) \rfloor < n$ selected pairs. If $N \geq Q$, then $|S| \geq Q/(d + 1)$. If $p > n + d\binom{N}{2}$, all N parameters can be made distinct.

Proof. For an admissible pole, assign

$$z_i = -R_i(\tau), \quad P_i(X) = \frac{R_i(X) - R_i(\tau)}{X - \tau}.$$

Then $\deg P_i < d$, and $f_0 + z_i f_1 - P_i = (w - R_i)/(X - \tau)$ on D , preserving the agreement set.

Two parameters collide only at a root of the nonzero polynomial $R_i - R_j$, which has degree at most d . Averaging over the Q admissible poles gives one with at most $dN(N - 1)/(2Q)$ colliding pairs. If the multiplicities of its distinct parameters are $m_1, \dots, m_v$, then

$$\sum_j m_j = N, \quad \sum_j m_j^2 \leq N + \frac{dN(N - 1)}{Q}.$$

Cauchy–Schwarz gives (10). Select one witness per parameter. If $N \geq Q$, the denominator is at most $(d+1)N$. Under the stronger field condition, the union of D and all pairwise root sets has size less than p , so a collision-free pole exists.

Any agreement of f_1 with a degree- $< d$ polynomial P_1 is a root of $(X - \tau)P_1(X) - 1$. This polynomial has degree at most d and is nonzero, as its value at τ is -1 . Thus even f_1 alone has at most d agreements with any codeword. Finally fix a parametrized codeword line $P_0 + zP_1$. Let $b \leq d$ be the number of coordinates where both coefficient words agree. At every other coordinate the two affine expressions agree for at most one parameter. If v selected witnesses lie on this line, counting agreements gives $v(t - b) \leq n - b$. The ratio $(n - b)/(t - b)$ increases with b , so $v \leq (n - d)/(t - d)$. $\square$

Theorem 3.3 (No uniform finite $n \exp(O_\rho(1/\eta))$ line threshold). *Fix rational $\rho \in (0, 1)$ and finite $c_1, c_2 > 0$. There are arbitrarily large prime-field Reed–Solomon codes of exact rate ρ on consecutive-integer domains that fail selected-witness line-decodability at concurrency n and threshold*

$$a = c_1 2^{c_2 H_2(\rho)/\eta} n, \quad \theta = 1 - \rho - \eta < r_E(\rho).$$

One can take $\eta = \Theta_\rho(1/\sqrt{n \log n})$ and $\log_2 p = \Theta_{\rho, c_2}(\sqrt{n \log n})$.

Proof. Let $d = \rho n$ be the target dimension. Choose $c^2 < 4H_2(\rho)$ and set $s = \lfloor c\sqrt{n/\log_2 n} \rfloor$, $t = d + s + 1$, and $\eta = (s + 1)/n$. Apply Lemma 2.1 to the source code of dimension $d + 1$. It supplies $N = 2^{\Omega_\rho(n)}$ degree- $\leq d$ polynomials with common agreement size t ; the one-dimensional change does not affect the estimate (3). Choose

$$A > \max\{H_2(\rho)/c, c_2 H_2(\rho)/c\}, \quad \log_2 p = A\sqrt{n \log_2 n} + O(1)$$

using Bertrand’s postulate. Eventually $N > p > n$. Lemma 3.2 gives at least $(p - n)/(d + 1)$ selected parameters, whose logarithmic count is $A\sqrt{n \log_2 n} - O(\log n)$. The proposed threshold has logarithm

$$\log_2(c_1 n) + c_2 H_2(\rho) \frac{n}{s + 1} = (c_2 H_2(\rho)/c + o(1)) \sqrt{n \log_2 n},$$

which is smaller. Every parametrized codeword line contains fewer than n selected witnesses. Finally $\eta \log_2 p \rightarrow cA > H_2(\rho)$ verifies Elias by the same entropy-continuity argument as above. $\square$

A finite line example. Take $p = 2^{127} - 1$, $n = 64$, target dimension $d = 32$, and agreement size $t = 34$. The first-moment sum 1071 has exactly

$$N = 8,634,334,451,040,106 > 2^{52}$$

34-subsets, as certified in Section 2.3. Their vanishing polynomials have two common leading terms, giving source polynomials of degree at most 32. The exact inequality $p > 64 + 32\binom{N}{2}$ makes all parameters distinct. Here $\eta = 1/32$, the threshold with $c_1 = c_2 = 1$ is 2^{38} , and each parametrized codeword line contains at most 16 selected witnesses. The radius is below Elias. In a smaller exhaustive check, $n = 12$, $d = 6$, $t = 8$, $p = 2^{31} - 1$ yield a moment class of size 33. The pole $\tau = 20$ gives 33 distinct parameters, and the maximum number of selected pairs on one parametrized codeword line is exactly 3, matching the lemma’s bound.

4 A fixed-gap obstruction that survives sublinear remainders

The preceding line theorem uses gaps that shrink with length. It therefore does not settle a proposed bound of the form $c_1 2^{c_2 H_2(\rho)/\eta} n + o(n)$ when the remainder is specified separately for each fixed gap. We now obtain a fixed-gap family.

There are two steps. First, keep one finite list example and replace each evaluation point by the same number of preimages under a power map. Degrees and agreement counts grow by the same factor, preserving both rate and gap. Second, use each nearby codeword on n distinct parameters of a word line. This forces a linear discrepancy from any proposed coefficient smaller than the list size.

Lemma 4.1 (A list example over the integers). *Fix rational $\rho \in (0, 1)$ and finite $c_1, c_2 \geq 1$. There are fixed integers $1 \leq k < t < m$, with $k/m = \rho$, distinct t -subsets $A_1, \dots, A_L \subset \{1, \dots, m\}$, and integer polynomials $W, G_1, \dots, G_L$ satisfying*

$$\deg G_i < k, \quad W(X) - G_i(X) = \prod_{a \in A_i} (X - a), \quad L > c_1 2^{c_2 H_2(\rho)/\eta}, \quad \eta = (t - k)/m.$$

The data is chosen before choosing a reducing prime.

Proof. Apply the moment construction over the integers. Equality of binomial moments implies equality of power sums over $\mathbb{Q}$, and Newton identities imply equality of the leading coefficients as integers. To see that a sufficiently large list exists without fixing a field, write $\rho = a/b$ and take $m = bu^3$, $k = au^3$, $s = u$, $t = k + s$. The logarithm of the lower bound in Lemma 2.1 is $bH_2(\rho)u^3 - O_\rho(u^2 \log u)$, whereas the logarithm of the proposed bound is $\log_2 c_1 + c_2 H_2(\rho)bu^2$. Choose a sufficiently large, now fixed, integer u and a class exceeding the proposed bound. Translating the original domain $\{0, \dots, m - 1\}$ by one, and substituting $X - 1$ in each polynomial, preserves the identities and avoids zero. $\square$

Lemma 4.2 (Replacing each point by a full power-map fiber). *Fix integer data as in Lemma 4.1. For every positive integer B and every bound T_0 , there is a prime $p > T_0$ and a domain $D_B \subset \mathbb{F}_p$ of size mB such that $x \mapsto x^B$ maps D_B onto $\{1, \dots, m\}$ with exactly B preimages per point. The polynomials $G_i(X^B)$ give distinct codewords in $\text{RS}(\mathbb{F}_p, D_B, kB)$, each agreeing with $W(X^B)$ on exactly tB coordinates.*

Proof. Let $K_B/\mathbb{Q}$ be the splitting field of $F_B(X) = \prod_{a=1}^m (X^B - a)$. The completely split case of Chebotarev's density theorem gives infinitely many primes that split completely in this finite Galois extension.¹ Exclude primes at most $\max\{T_0, m, B\}$ and primes dividing the nonzero discriminant of F_B . At any remaining such prime, all roots of F_B reduce to distinct elements of $\mathbb{F}_p$. Let D_B be these roots. The roots of each $X^B - a$ form one fiber of size B , and the fibers are disjoint.

Now $\deg G_i(X^B) \leq (k - 1)B < kB$, and

$$W(X^B) - G_i(X^B) = \prod_{a \in A_i} (X^B - a).$$

The difference vanishes on exactly the t selected fibers. The codewords remain distinct: for $A_i \neq A_j$, a point in a fiber indexed by $A_i \setminus A_j$ makes exactly one of the two displayed products zero. No further coefficient-preservation condition is needed. $\square$

Only the existence of arbitrarily large completely split primes is used; we require no bound on the least such prime. The construction may therefore use fields much larger than those in Theorem 2.2.

¹See J. S. Milne, *Algebraic Number Theory*, Theorem 8.31 and Corollary 8.32, [course notes](#). Only infinitude of completely split primes is used.

Lemma 4.3 (A list forces a selected-witness threshold). *Let $C \subset F^n$ be a nonzero linear code. Suppose a word w has L distinct codewords $q_1, \dots, q_L$ within distance θ , where $L \leq n$ and $|F| \geq Ln$. There is a nonconstant affine word line with Ln distinct parameters and a selected nearby codeword at each, such that no parametrized codeword line contains more than n selected pairs. Thus line-decodability with concurrency n fails at every positive threshold smaller than Ln .*

Proof. Choose nonzero $v \in C$ and disjoint sets $S_1, \dots, S_L \subset F$, each of size n . Put $f(z) = w + zv$ and select $p_z = q_i + zv$ for $z \in S_i$. Translation preserves distance, so every selected codeword is nearby. Fix $P_0, P_1 \in C$. If $P_1 = v$, the equality $P_0 + zP_1 = q_i + zv$ holds only for the branch with $P_0 = q_i$, if any, and that branch has n parameters. If $P_1 \neq v$, the equation $P_0 - q_i = z(v - P_1)$ has at most one solution per branch, giving at most $L \leq n$ selected pairs. The argument compares pairs at the same z ; it does not require the unparametrized images of the lines to be disjoint. $\square$

Theorem 4.4 (Failure at a fixed gap with any sublinear remainder). *For every rational $\rho \in (0, 1)$ and finite $c_1, c_2 \geq 1$, there are a fixed gap $\eta \in (0, 1 - \rho)$, an integer $L > c_1 2^{c_2 H_2(\rho)/\eta}$, and a sequence of prime-field Reed–Solomon codes of lengths $n_B \rightarrow \infty$ and exact rate ρ , all with $\theta = 1 - \rho - \eta$ strictly below their characteristic-based Elias radii, such that the following holds. Each code admits Ln_B selected nearby witnesses on a nonconstant affine word line, but no parametrized codeword line contains more than n_B selected pairs. Consequently, for every sequence $R_B = o(n_B)$, all sufficiently large codes fail line-decodability with concurrency n_B and threshold*

$$a_B = c_1 2^{c_2 H_2(\rho)/\eta} n_B + R_B. \quad (11)$$

The code family is chosen independently of the remainder sequence.

Proof. Fix the integer example from Lemma 4.1; hence m, k, t, L, η are now constants. Let B grow with $n_B = mB \geq L$. For each B , choose the least completely split prime from Lemma 4.2 satisfying

$$p_B > \max\{Ln_B, m, 2^{1/\eta}\}$$

and the separability condition in that lemma. This fixes a prime sequence without reference to R_B . The lifted code has rate $k/m = \rho$ and contains a list of size L at distance exactly $1 - t/m = \theta$. It is nonzero, and the hypotheses of Lemma 4.3 hold. Moreover

$$H_{p_B}(\theta) \leq \theta + 1/\log_2 p_B < \theta + \eta = 1 - \rho,$$

which verifies the Elias condition. Put $A = c_1 2^{c_2 H_2(\rho)/\eta}$. Since $L - A > 0$ is fixed, eventually $R_B < (L - A)n_B$, so $a_B < Ln_B$. Lemma 4.3 contradicts line-decodability at this threshold. $\square$

The gap can depend on the proposed constants, but is fixed along the resulting family. Even a remainder chosen separately for that family cannot absorb its positive linear discrepancy. This last construction concerns the arbitrary selection of nearby witnesses: it has correlated agreement, since (q_i, v) agrees jointly with (w, v) wherever q_i agrees with w . Also, its selected count Ln never exceeds $\ell_C(\theta)n$. Thus the obstruction rules out the proposed quantitative coefficient, not a line threshold expressed using a valid bound on the actual maximum list size.

5 Small lists do not bound the number of nearby line points

The previous constructions produced large lists. We now show a different obstruction: even a code with small lists can have an affine line with exponentially many nearby points, each with a

unique nearby codeword. The degree argument is elementary. To keep the code's *maximum* list size small, we choose its evaluation domain using a generic-rank theorem for Reed–Solomon agreement matrices and a finite-field union bound. The underlying line is the consecutive-power construction of Ben-Sasson, Carmon, Haböck, Kopparty, and Saraf [11, Theorem 7.1], also used in [10]. Here we choose a domain on which that line coexists with a small global maximum list size at the target radius.

Recall that $\ell_C(\theta)$ is the maximum number of codewords within relative Hamming distance θ of any received word. A selected pair (z, p_z) lies on a parametrized codeword line if $p_z = P_0 + zP_1$ for two fixed codewords $P_0, P_1 \in C$; the scalar z must be the original parameter. Correlated agreement on t coordinates means that $P_0 = f_0$ and $P_1 = f_1$ simultaneously on those coordinates.

Theorem 5.1 (An actual-list-size separation). *Fix any rational rate $0 < \rho < 1$. For every sufficiently large n with $k = \rho n$ integral, there are a prime p , an n -point domain $D \subset \mathbb{F}_p$, and the rate- ρ code $C = \text{RS}(\mathbb{F}_p, D, k)$ with the following properties at*

$$\theta = 1 - \rho - \frac{1}{n}.$$

- (i) *The actual maximum list size satisfies $\ell_C(\theta) = n - k - 1$.*
- (ii) *An affine line $f_0 + zf_1$ has exactly $M = \binom{n}{k+1}$ parameters within distance θ of C . Each has a unique nearby codeword p_z .*
- (iii) *Every parametrized codeword line contains at most $n - k$ of these selected pairs (z, p_z) , and this bound is attained. The pair (f_0, f_1) has no correlated agreement on $k + 1$ coordinates.*
- (iv) *The radius θ lies strictly below the characteristic- p Elias radius.*

Consequently, along this family, line-decodability with concurrency n fails at threshold $\ell_C(\theta)n + R_n$ for every $R_n = o(n)$.

Proof. Choose the domain. For sufficiently large n , we have $k \geq 2$ and $a = n - k \geq 2$; take a prime $p \geq n + k2^{10an}$. Sample n distinct domain points uniformly from $\mathbb{F}_p$. We impose three conditions on this domain.

First, put $L = a - 1 \geq 1$. We use the generic-matrix argument underlying [4, Sections 2.2–2.4 and 3.1], with an explicit finite-field union bound. If $L + 1$ distinct codewords lie in a radius- θ ball, their total (unnormalized) distance from its center is at most $(L + 1)(a - 1) = La = L(n - k)$. The non-strict inequality in [4, Lemma 2.3] permits equality and implies that some subcollection of t codewords, $2 \leq t \leq L + 1$, has a k -weakly-partition-connected agreement hypergraph. Its vertices are these codewords, and its i th edge records which agree with the center at coordinate i . Weak partition connectivity means that every partition $\mathcal{P}$ of the vertices satisfies $\sum_i \max\{|\mathcal{P}(e_i)| - 1, 0\} \geq k(|\mathcal{P}| - 1)$, where $\mathcal{P}(e_i)$ denotes the parts meeting e_i .

The associated reduced intersection matrix encodes equalities of polynomial values within each edge, using a spanning star, with the coefficient block for the last polynomial removed. It has $(t - 1)k$ columns and entries of degree at most $k - 1$ in the formal domain points. It has full symbolic column rank by [4, Theorem 2.11]. However, at a domain realizing the proposed subcollection, the coefficient differences from its last codeword give a nonzero kernel vector. Thus a fixed nonzero full-column minor must vanish there. This minor has total degree at most $(t - 1)k(k - 1) \leq Lk^2$.

There are at most $\sum_{t=2}^{L+1} 2^{tn} \leq 2^{(L+2)n}$ such labeled hypergraphs. For independent uniform domain points, Schwartz–Zippel and a union bound give failure probability at most $2^{(L+2)n} Lk^2/p$.

Since $p > n^2$, the probability that those points are distinct is greater than $1/2$. Conditioning on distinctness therefore gives failure probability at most

$$\frac{2^{(L+2)n+1} L k^2}{p} < 2^{-2an}.$$

For the last inequality, use $L = a - 1$, $p > k2^{10an}$, $2 \leq a < n$, and $1 + \log_2((a-1)k) < (7a-1)n$ for $n \geq 3$. Consequently, except with probability 2^{-2an} , every radius- θ list has size at most $a - 1$.

Second, require all $(k+1)$ -subsets of the domain to have distinct sums. For two different index sets T, U , condition on every domain point except one indexed by $T \setminus U$. Equality of the two sums fixes at most one value of this remaining point, among $p - n + 1$ possibilities. The probability of any collision is therefore at most $\binom{M}{2}/(p - n + 1)$. Our choice of p gives

$$2^{-2an} + \frac{\binom{M}{2}}{p - n + 1} < 1.$$

Indeed, $M \leq 2^n$ and $p - n + 1 > k2^{10an}$ make the second failure probability less than $2^{2n-1-10an} \leq 2^{-2an}$. Finally, impose the matching-list construction from Proposition 5.2 below, simultaneously at all integer radii strictly below $n - k$. Its proof bounds failure by $(a-1)^2/[2(p-n+1)] < 2^{-2an}$. Together with the preceding two bounds, total failure is less than $3 \cdot 2^{-2an} < 1$. Fix a domain satisfying all three conditions. The matching construction in particular gives $\ell_C(\theta) \geq a - 1$, so the bound at our target radius is exact.

Construct the nearby points. For each $(k+1)$ -subset $T \subset D$, put

$$s_T = \sum_{a \in T} a, \quad V_T(X) = \prod_{a \in T} (X - a).$$

Evaluate the affine family

$$f_z(X) = X^{k+1} - zX^k$$

on D . At parameter $z = s_T$, choose

$$p_{s_T}(X) = X^{k+1} - s_T X^k - V_T(X).$$

Its two leading coefficients cancel, so $\deg p_{s_T} < k$. Its difference from f_{s_T} vanishes exactly on T , giving distance $1 - (k+1)/n = \theta$.

Conversely, if a polynomial P of degree less than k agrees with f_z on at least $k+1$ coordinates, then $f_z - P$ is monic of degree $k+1$ and must equal V_T for its root set $T \subset D$. Comparing the coefficient of X^k gives $z = s_T$. Distinct subset sums therefore identify both T and P uniquely. There are exactly M nearby parameters, each with one nearby codeword.

Bound concurrency. Suppose $P_0 + zP_1$ contains m selected pairs, and let b count the coordinates where both $X^{k+1} = P_0(X)$ and $-X^k = P_1(X)$. Since $\deg P_1 < k$, the second equality holds on at most k coordinates; hence $b \leq k$, which already rules out the claimed common agreement. At those b coordinates all m parameters may agree. At every other coordinate two distinct affine functions of z agree at most once. Counting the $k+1$ agreements of each selected witness yields

$$m(k+1) \leq bm + n - b, \quad m \leq \frac{n-b}{k+1-b} \leq n-k.$$

The bound is attained by fixing a k -subset $A \subset D$ and writing $V_A(X) = \prod_{a \in A} (X - a)$ and $s_A = \sum_{a \in A} a$. The codeword line

$$[X^{k+1} - (X + s_A)V_A(X)] + z[V_A(X) - X^k]$$

has both coefficient polynomials of degree less than k and contains the selected witness for each $T = A \cup \{a\}$ with $a \in D \setminus A$, at its parameter $z = s_A + a$. There are $n - k$ such parameters.

Check the radius and threshold. The entropy estimate $H_p(\theta) \leq \theta + 1/\log_2 p < 1 - \rho$ places θ strictly below the Elias radius. Finally, $M = 2^{H_2(\rho)n - O_\rho(\log n)}$, whereas $\ell_C(\theta)n \leq (n - k - 1)n = O_\rho(n^2)$. Thus $M > \ell_C(\theta)n + R_n$ eventually for every $R_n = o(n)$, without the required concurrency. $\square$

Attribution of the exact profile. The upper bound below follows from [5, Theorem 1.5]. For the lower bound, Roth's Theorem 4 [6] suffices whenever $a - 2 \leq \binom{k+1}{2}$. Indeed, set $L_s = \lfloor (a - 1)/s \rfloor$. If $L_s \geq 2$, put $L = L_s - 1$ and write $a = (L + 1)u + v$ with $1 \leq v \leq L + 1$. Then $u \geq 1$ and $L \leq a - 2 \leq \binom{k+1}{2} \leq \binom{k-1+u+v}{k-1}$. If the maximum list size at radius $(a - s)/n$ were at most L , Roth's bound would imply $a - s \leq La/(L + 1)$, contrary to $L_s s \leq a - 1$. The case $L_s = 1$ is immediate. The sufficient condition holds eventually at every fixed rate in $(0, 1)$. Thus the fixed-rate exact profile is a consequence of published results. The proof below supplies a direct lower construction for every finite $k \geq 2$ and uses [4] for the stated finite-field probability.

Proposition 5.2 (Exact generic list sizes at every integer radius below $n - k$). *Let $k \geq 2$, $a = n - k \geq 2$, and let $p \geq n + k2^{10an}$ be prime. For a uniformly random ordered set of n distinct evaluation points, with probability greater than $1 - 2^{1-2an}$, the Reed–Solomon code C simultaneously satisfies*

$$\ell_C\left(\frac{a-s}{n}\right) = \left\lfloor \frac{a-1}{s} \right\rfloor \quad (1 \leq s < a).$$

At radius zero the maximum list size is one. The domain in Theorem 5.1 can be chosen to have this entire profile, as well as distinct sums of its $(k + 1)$ -subsets.

Proof. Upper bounds from one rank condition. Use the good-domain event in the proof of Theorem 5.1: every k -weakly-partition-connected agreement hypergraph on at most a vertices has a reduced intersection matrix of full column rank. Its failure probability is less than 2^{-2an} . For each s , put $L_s = \lfloor (a - 1)/s \rfloor$. If $L_s + 1$ distinct codewords are each at distance at most $a - s$ from one word, their total distance is at most

$$(L_s + 1)(a - s) \leq L_s a,$$

since $(L_s + 1)s \geq a$. Lemma 2.3 of [4] then produces one of the excluded hypergraphs on at most $L_s + 1 \leq a$ vertices. Thus the same rank condition supplies all upper bounds simultaneously.

Matching lists from a chain of affine polynomials. Set aside $k - 2$ coordinates H and put $F(X) = \prod_{h \in H} (X - h)$. There remain $a + 2$ coordinates. Fix s and write $L = L_s$. Choose two anchors A, B , another s coordinates, and a seed z_0 . Give the anchors and the other s coordinates received value zero, and give z_0 value one. The zero polynomial already has $s + 2$ agreements. For each $1 \leq j \leq L - 1$, choose a fresh set Z_j of s coordinates and designate one of them z_j . Let $A_j = A$ for odd j and $A_j = B$ for even j , and define

$$c_j = \frac{w(z_{j-1})}{z_{j-1} - A_j}, \quad q_j(X) = c_j(X - A_j), \quad w(x) = q_j(x) \quad (x \in Z_j).$$

This uses $s + 3 + (L - 1)s = Ls + 3 \leq a + 2$ coordinates. Give any remaining coordinates value zero. All denominators and slopes are nonzero, and q_j agrees with w at the anchor A_j , the preceding point z_{j-1} , and all s points of Z_j . Each candidate therefore has at least $s + 2$ agreements.

It remains to ensure that the candidates are distinct. Candidates with different anchors have different roots. For $i < j$ with the same anchor, the equality $q_i = q_j$ is equivalent to

$$\prod_{r=i}^{j-1} \frac{z_r - A_r}{z_r - A_{r+1}} = 1.$$

Condition on every domain coordinate except z_i . All factors but one are fixed and nonzero, so the equality takes the form $C(z_i - A_i)/(z_i - A_{i+1}) = 1$ with $C \neq 0$. Because the anchors differ, this equation has at most one solution. Under distinct sampling its probability is therefore at most $1/(p - n + 1)$. There are at most $L^2/4$ pairs with the same anchor, so candidate repetition has probability at most $L^2/[4(p - n + 1)]$. On its complement, multiplication by F gives L distinct polynomials of degree less than k . Set the full received word to zero on H and to $F(x)w(x)$ elsewhere. Each candidate has at least $(k - 2) + (s + 2) = k + s$ agreements, proving the matching lower bound.

Choose one domain for all radii. The sum of the lower-construction failure probabilities is at most

$$\frac{1}{4(p - n + 1)} \sum_{s=1}^{a-1} L_s^2 < \frac{(a - 1)^2}{2(p - n + 1)} < 2^{-2an},$$

using $L_s \leq (a - 1)/s$ and $\sum_{s \geq 1} s^{-2} < 2$. Together with the single rank-condition failure bound, this proves the probability assertion. Requiring distinct $(k + 1)$ -subset sums adds less than 2^{-2an} to the failure probability, as in the preceding proof; the total remains less than one. $\square$

The upper bounds here use the known generic-rank theorem; the lower bounds are explicit interpolation constructions. The domain is generic and the field bound is large. The assertion does not give this list profile on arbitrary prescribed evaluation domains. Dimension $k = 1$ is a real exception: at agreement $1 + s$, the constant code has maximum list size $\lfloor n/(s + 1) \rfloor$.

Corollary 5.3 (Extension fields and interleaving). *The domain in Proposition 5.2 has the same exact list profile over every extension of its base field and for every finite interleaving order. Interleaved distance is measured by columns: a coordinate agrees only when all components agree. No further exceptional-domain event is needed.*

Proof. A nonzero full-column minor over the base field remains nonzero over an extension. For interleaved words, agreement of entire columns gives the same agreement hypergraph as in the scalar upper-bound proof. Apply its full-rank matrix to each component of the polynomial coefficient differences. Each component difference vanishes, so two allegedly distinct tuples would coincide. The scalar lower list embeds in one component, with zero in the others, and remains a list over every extension field. This proves equality. The argument does not apply to entrywise distance averaged over all components. $\square$

The companion `checks/astra_rs_all_surplus_check.py` verifies 3,654 large-field parameter fixtures on 378 domains, each reused for all surpluses, including nondivisible surpluses and cases with $k < 2s$. It also checks 450 small-field fixtures, retaining 55 with repeated candidate lines, and enumerates the full lists at eight constructed centers. The 79,079 allocation and probability checks, and 3,003 combined lower-failure checks, pass. Separately, `checks/actual_list_global_small_check.py` exhausts all center classes modulo the code in four small dimension-two examples. Three structured domains have a larger list than the generic profile, confirming the need for the domain qualification.

Sharpness of the global list bound at rates at least one-half. Write $a = n - k \geq 2$. If $k \geq a - 2$, every n -point Reed–Solomon domain has a list of size at least $a - 1$ at radius $(a - 1)/n$. Indeed, set $L = a - 1$ and $h = k - a + 2 \geq 0$, and partition the domain into H , $\{\alpha_1, \dots, \alpha_L\}$, and $\{\beta_1, \dots, \beta_L\}$, where $|H| = h$. Let $F(X) = \prod_{x \in H} (X - x)$ and

$$Q_i(X) = F(X) \left[X^{L-1} - \prod_{j \neq i} (X - \alpha_j) \right] \quad (1 \leq i \leq L).$$

The leading terms cancel, so each has degree at most $h + L - 2 = k - 1$ (for $L = 1$, the sole polynomial is zero). For $i \neq j$, their values at α_i differ by the nonzero quantity $F(\alpha_i) \prod_{r \neq i} (\alpha_i - \alpha_r)$, so they are distinct. Choose a received word equal to zero on H , to $F(\alpha_j) \alpha_j^{L-1}$ at α_j , and to $Q_i(\beta_i)$ at β_i . Each Q_i agrees on H , on every α_j with $j \neq i$, and on β_i , giving $h + L = k + 1$ agreements. This supplies the matching lower bound on every domain in this range. The chain construction in Proposition 5.2 supplies a matching lower bound on a generic domain for every $k \geq 2$.

A sharper collision estimate. For the $s = 1$ chain, the probability of a repeated line is at most

$$\frac{\lfloor \max(a - 5, 0)^2 / 4 \rfloor}{p - n + 1}.$$

For a same-anchor pair $i < j$, condition on all domain points except z_i . Equality of the two slopes has the form $C(z_i - A_i)/(z_i - A_{i+1}) = 1$ with fixed $C \neq 0$. If $C = 1$, equality is impossible; otherwise it fixes at most one of the $p - n + 1$ remaining values of z_i . Pairs with $j - i = 2$ cannot collide: their cleared equation is, up to sign, $(A - B)(z_{i+1} - z_i) = 0$. Among $a - 2$ alternating lines, the number of same-anchor pairs separated by at least four steps is $\lfloor \max(a - 5, 0)^2 / 4 \rfloor$, proving the estimate. In particular, the chain is distinct on every domain for $a \leq 6$. The independent checker `checks/actual_list_chain_conditional_audit.py` exhausts 36,200 allowed values across 1,400 conditional pair cases over $\mathbb{F}_{37}$, verifying the one-solution bound and all 720 gap-two exclusions in those fixtures.

The gap in Theorem 5.1 is $\eta = 1/n$. It therefore rules out a *uniform*, familywise threshold based on the actual list size; it does not rule out an asymptotic statement made separately at each fixed positive gap, with arbitrary dependence of its remainder on that gap. This differs from the separate fixed-gap construction, which defeats a prescribed numerical list coefficient. The present construction instead has provably small actual lists. Its ingredients are the vanishing-polynomial construction and a domain chosen to satisfy the generic-rank guarantee simultaneously with distinct subset sums.

5.1 A general upper bound and a sharp endpoint

The affine-line count in Theorem 5.1 attains a general upper bound at the same radius. The upper bound follows by puncturing to the unique-decoding regime and applying the correlated-agreement theorem of Ben-Sasson et al. [2, Theorems 1.2 and 1.4]. We include the argument to explain the extremality of the example; no novelty is claimed for this corollary of the known theorem. Stronger bounds are known in parts of the unique-decoding and Johnson regimes [11, Corollary 1.4 and Theorem 1.5]; the elementary bound below is useful for its general range and its sharp endpoint.

Proposition 5.4 (An unconditional puncturing bound). *Let $C = \text{RS}(F, D, k)$, where F is finite, $|D| = n$, and $a = n - k$. Let r be an integer with $a/2 \leq r < a$, and put $s = a - r$ and $d = 2r - a$. Suppose that $f_0, f_1 \in F^D$ have no correlated agreement with degree-below- k polynomials on $n - r$ coordinates. Then the total number of nearby witness pairs satisfies*

$$\sum_{z \in F} |\{P : \deg P < k, \text{dist}(f_0 + z f_1, P|_D) \leq r/n\}| \leq \left\lfloor (n - d) \frac{\binom{n}{d}}{\binom{r}{d}} \right\rfloor = \left\lfloor s \frac{\binom{n}{d+1}}{\binom{r}{d+1}} \right\rfloor.$$

Proof. Delete any d coordinates. The resulting code has length $n - d$, dimension k , and redundancy $a - d = 2s$. Correlated agreement on $(n - d) - s = n - r$ remaining coordinates would also give correlated agreement in the original domain, contrary to the hypothesis. The unique-radius theorem

therefore permits at most $n - d$ parameters whose punctured words have at most s errors. At each parameter the nearby polynomial is unique, because the punctured code has minimum distance $2s + 1$. Its relative radius is $s/(n - d) = (1 - k/(n - d))/2$, the included endpoint of the cited unique-radius bound.

For each nearby witness pair (z, P) in the original code, let its error support be E , of size at most r , and extend E to an r -element coordinate set R . Deleting any d -subset of R leaves at most $r - d = s$ errors. Thus this pair is counted by at least $\binom{r}{d}$ punctures. Restriction preserves distinct polynomials because $n - d \geq k$. There are $\binom{n}{d}$ punctures, each counting at most $n - d$ witness pairs. Double counting proves the first bound. The second follows from $r - d = s$ and the adjacent binomial identities. The case $d = 0$ uses the empty puncture. $\square$

At $r = a - 1$, the bound is exactly $\binom{n}{k+1}$. Theorem 5.1 attains it, while its global maximum list size is only $n - k - 1$. Consequently, among affine lines without correlated agreement at this radius, the number of nearby parameters in that theorem is as large as possible. This extremality concerns an algebraic count, not an algorithm for producing its witnesses.

Proposition 5.5 (Pairs with unexplained agreement sets). *Retain the parameters of Proposition 5.4, but allow f_0, f_1 to have correlated agreement. For a nearby pair (z, P) , let $S = \{x \in D : P(x) = f_0(x) + zf_1(x)\}$ be its entire agreement set. The number of such pairs for which no two degree-below- k polynomials jointly explain f_0, f_1 on S is at most*

$$\left\lfloor (n - d) \frac{\binom{n}{d}}{\binom{r}{d}} \right\rfloor.$$

Proof. First consider a length- N puncture of redundancy $2s$. If its coefficient words have no correlated agreement on $N - s$ coordinates, the unique-radius theorem and uniqueness of the nearby polynomial bound all nearby pairs by N . Otherwise, choose a jointly agreeing codeword path $Q_0 + zQ_1$, and let J be its full coefficient agreement set. This path is nearby at every parameter and hence is the unique nearby polynomial. Outside J , an accidental agreement can occur at at most one parameter per coordinate. There are at most $N - |J| \leq s$ such parameters. At every other parameter the entire agreement set is J and is jointly explained. Thus at most N pairs in any puncture have unexplained entire agreement sets.

For an original pair with an unexplained set S , interpolate the two coefficient words on any k positions of S . At least one further position fails one of these interpolants. The resulting $(k + 1)$ -set $K \subset S$ is itself not jointly explainable. Its complement has size $a - 1 \geq r$, so the error support $E = D \setminus S$ can be extended to an r -set $R \subseteq D \setminus K$. Every d -subset of R leaves at most s errors after puncturing and preserves K . The punctured agreement set therefore remains unexplained. Each original pair is counted in at least $\binom{r}{d}$ punctures, each of which counts at most $n - d$ such pairs. Restriction is injective on the polynomial witnesses, so double counting gives the bound. $\square$

Corollary 5.6 (Polynomial curves). *Retain a, r, s, d from Proposition 5.4, and let $f(z) = \sum_{j=0}^e z^j f_j$ for $f_j \in F^D$ and $e \geq 0$. The number of nearby pairs (z, P) whose entire agreement set cannot be jointly explained by degree-below- k polynomials for all $e + 1$ coefficient words is at most*

$$\left\lfloor e(n - d) \frac{\binom{n}{d}}{\binom{r}{d}} \right\rfloor.$$

If no such joint agreement exists on $n - r$ coordinates, the same bound holds for all nearby pairs, summed over parameters.

Proof. This is an elementary consequence of the polynomial-curve unique-radius theorem [2, Theorem 6.1]. On a length- N puncture of redundancy $2s$, that theorem bounds nearby parameters by eN unless the coefficient words jointly agree with codewords on $N - s$ positions. Uniqueness makes this a witness bound. In the joint-agreement case, choose the corresponding codeword curve and its full coefficient agreement set J . It is nearby and unique at every parameter. Each coordinate outside J contributes at most e roots of its nonzero discrepancy polynomial. Thus at most $e(N - |J|) \leq es$ parameters have an agreement set larger than J ; all remaining sets are jointly explained.

For each original unexplained support, interpolate every coefficient word on the same k positions and retain one additional position where an interpolant fails. This gives an unexplained $(k + 1)$ -set, independent of e . The protected-support counting argument in Proposition 5.5 now applies with eN in place of N , yielding the displayed bound. For $e = 0$, every nearby support is explained by its nearby polynomial, so the count is zero. The root bound uses formal polynomials and requires no assumption that $e < |F|$. $\square$

The affine bound is also n at $2r = a$. Between the two endpoints it can be much larger than the actual maximum list size times n ; at a fixed gap beyond unique decoding it is generally exponential in n . It therefore leaves the fixed-gap question open. The exact checker `checks/verify_puncturing_bound.py` verifies the parameter and support-count identities and enumerates all 257 parameters on each of five small lines attaining the one-coordinate-gap bound. The independent `checks/astra_syndrome_determinants_check.py` checks the witness and entire-agreement-set bounds on 378 affine lines, including lines with correlated agreement, by exact decoding and 572,226 codeword-distance evaluations. For the curve corollary, `checks/verify_curve_puncturing.py` enumerates all field parameters and codewords on 300 deterministic small curves, including 60 with formal degree at least the field size. All 466,140 distance checks pass; the fixtures include 421 unexplained witness pairs on curves with some joint agreement.

6 Finite examples on the norm-one circle

The interval construction shows that a uniform list bound fails when the field and block length vary. We now give finite examples over one fixed prime, $p = 2^{31} - 1$. The domains have multiplicative structure, and the construction uses that structure instead of small integer moments. The two methods share a simpler underlying idea: many root polynomials can have the same outer coefficients while differing in their middle coefficients.

6.1 Why the circle helps

Let $i^2 = -1$ in $F_2 = \mathbb{F}_{p^2}$. The norm-one circle is the cyclic group

$$U = \{u \in F_2^\times : u^{p+1} = 1\}.$$

Every element can be written $u = x + iy$ with $x, y \in \mathbb{F}_p$ and $x^2 + y^2 = 1$. On this group $u^p = u^{-1}$. Thus conjugating a polynomial's coefficients and reversing its exponents are closely related operations. For $n \mid p + 1$, write μ_n for the subgroup of size n . Our evaluation domains are μ_n and its rotations $g\mu_n$.

For even $K = 2m$, consider the dimension- K circle space

$$C_K = \{A(x) + yB(x) : \deg A < m, \deg B < m\}.$$

Its coefficients may lie in $\mathbb{F}_p$ or in an extension, as specified. In the coordinate u , the space contains all Laurent polynomials in the interior band $-(m - 1), \dots, m - 1$, with the appropriate coefficient

field condition. A Laurent polynomial simply allows negative powers of u ; these are well defined because every evaluation point is nonzero. The interior band leaves room to divide out two common zeros without leaving C_K .

Write $n = MB$. The power map $u \mapsto u^B$ sends μ_n onto μ_M , with B points in each fiber. To choose agreement positions, choose an h -element subset $A \subseteq \mu_M$ containing 1, and form

$$F_A(Y) = \prod_{a \in A} (Y - a) = \sum_{j=0}^h f_{A,j} Y^j.$$

The roots have norm one, so, with $c = f_{A,0}$,

$$f_{A,j} = c f_{A,h-j}^p.$$

Fixing c and r leading coefficients therefore also fixes r trailing coefficients. There are $\binom{M-1}{h-1}$ possible subsets, but at most Mp^{2r} choices of this coefficient data. Some class has at least

$$\left\lceil \frac{\binom{M-1}{h-1}}{Mp^{2r}} \right\rceil$$

members. Subtracting their common outer terms leaves distinct low-degree polynomial parts. Agreement occurs on the selected fibers, so the count of roots in μ_M becomes a count of coordinates in μ_n .

Two additional steps yield the two results below. For a base-field list, one rescales the polynomial parts to make their values lie in $\mathbb{F}_p$. The common anchor $1 \in A$ then permits division at the two circle points $u = 1, -1$. For an affine family, one instead divides by $Y - \tau$, where $\tau \in \mathbb{F}_{p^4} \setminus \mathbb{F}_{p^2}$. As in Lemma 3.2, this produces a scalar label for each polynomial part and a direction word with too little agreement to permit correlated agreement. A collision count selects a pole with many distinct labels. Appendix A gives both arguments in full.

6.2 Two constructions at the same parameters

Theorem 6.1 (Finite circle list and affine-family bounds). *Let $p = 2^{31} - 1$, $E = \mathbb{F}_{p^4}$, and*

$$n = 2^{26}, \quad K = 2^{25}, \quad \rho = \frac{1}{2}, \quad \eta = \frac{33}{1024} - \frac{2}{n}, \quad T = (\rho + \eta)n, \quad \theta = 1 - \rho - \eta.$$

On $D = \mu_n$, and on any rotated domain $g\mu_n \subseteq U$, the following statements hold.

(i) *There is an $\mathbb{F}_p$ -valued received word with more than*

$$2^{40}$$

distinct codewords from the base-field circle space C_K agreeing with it in at least T positions.

(ii) *There are words $f_0, f_1 \in E^D$ with no correlated agreement at threshold T with the extension-field circle code C_K , but more than*

$$2^{74}$$

distinct scalars $z \in E$ for which $f_0 + zf_1$ has a codeword agreeing in at least T positions. Witnesses can be selected so that every parametrized affine codeword graph contains at most fifteen selected pairs (z, P_z) .

The radius θ is strictly below the characteristic- p Elias radius at rate $1/2$.

Proof. For (i), apply Theorem A.2 with $M = 2048$, $d = 1024$, $r = 32$, $h = 1090$, and $B = 2^{15}$. For (ii), apply Theorem A.4 with $M = 1024$, $r = 15$, $h = 545$, and $B = 2^{16}$. Both choices give $K = n/2$ and the same agreement count $T = hB - 2$. The exact coefficient counts and collision bound are evaluated in (20) and Section A.1.2. Corollary A.9 gives the selected-graph bound. The strict Elias comparison has the rational certificate in Section A.1.3. $\square$

These are two different received-word constructions with the same length, rate, and radius. Neither the theorem nor its proof requires enumerating the very large coefficient classes. The formulas and integer lower bounds are explicit; the choice of a sufficiently large class, and of a suitable pole in (ii), is guaranteed by counting.

Transfer to ordinary Reed–Solomon codes. The second construction also gives an ordinary Reed–Solomon example. Remove the two anchor coordinates and use the stereographic coordinate $t = y/(1 + x) \in \mathbb{F}_p$. Multiplication by the nonzero weight $(1 + t^2)^{m-1}$ identifies the interior Laurent band with ordinary polynomials of degree at most $K - 2$. The resulting code has

$$n' = n - 2, \quad k' = K - 1, \quad \rho' = \frac{1}{2}, \quad \eta' = \frac{T}{n - 2} - \frac{1}{2} = \frac{33 \cdot 2^{16} - 1}{n - 2}.$$

It has the same scalar labels and at least T agreements per selected witness, with no correlated agreement at threshold T . Theorem A.10 proves that the coordinate change and scaling preserve these equalities. The new gap is η' , not η : puncturing preserves the number of certified agreements but changes the block length. Comparisons for this Reed–Solomon code must use n' and η' .

6.3 Stronger finite examples at other rates

The preceding examples use rate $1/2$ to make the two constructions directly comparable. Neither construction is restricted to that rate. In the list construction, choose any even packet dimension d and put $h = d + 2r + 2$. The resulting rate is d/M ; the gap remains $(2r + 2)/M - 2/n$. In particular, no coefficient reserve is needed for the following larger list.

Corollary 6.2 (A quarter-rate base-field list). *Let $p = 2^{31} - 1$, $n = 2^{26}$, $K = 2^{24}$, and*

$$\rho = \frac{1}{4}, \quad \eta = \frac{1}{32} - \frac{2}{n}.$$

On μ_n and its rotations, some $\mathbb{F}_p$ -valued word has more than 2^{43} codewords in C_K agreeing on at least $(\rho + \eta)n$ coordinates. The radius $1 - \rho - \eta$ is strictly below the characteristic- p Elias radius. If L denotes the certified list size, then

$$\frac{L}{2^{H_2(\rho)/\eta}} > 200,000.$$

Proof. Use Theorem A.2 with $M = 64$, $B = 2^{20}$, $d = 16$, $r = 0$, and $h = 18$. The count is $\lceil \binom{63}{17}/64 \rceil$. Appendix D.2 gives exact certificates for the count, radius, and displayed comparison. $\square$

Corollary 6.3 (An eighth-rate line without correlated agreement). *Let $p = 2^{31} - 1$, $E = \mathbb{F}_{p^4}$, $n = 2^{26}$, $K = 2^{23}$, and*

$$\rho = \frac{1}{8}, \quad \eta = \frac{5}{256} - \frac{2}{n}, \quad T = (\rho + \eta)n.$$

On μ_n and its rotations, there are words $f_0, f_1 \in E^D$ with no correlated agreement at threshold T with C_K , but more than 2^{75} distinct scalars z for which $f_0 + zf_1$ has a codeword agreeing on at least T coordinates. If J is the certified scalar count, then

$$\frac{J}{n2^{H_2(\rho)/\eta}} > 2^{22}.$$

The radius is strictly below the characteristic- p Elias radius. Witnesses can be selected so that every parametrized affine codeword graph contains at most 55 selected pairs.

Proof. Apply Theorem A.4 with $M = 256$, $B = 2^{18}$, $r = 1$, and $h = 37$. Its polynomial degree is $d = 33$, so its collision factor is $d - 1 = 32$. The second word agrees with every outer-band codeword at at most $a_0 = K + B = 33B < T = 37B - 2$ coordinates. Lemma A.8 bounds the number of selected pairs on one graph by

$$\left\lfloor \frac{223B}{4B - 2} \right\rfloor = 55.$$

The exact scalar count, Elias comparison, and ratio certificate are in Appendix D.2. $\square$

The small packet group explains the improvement in the list construction: only 64 possible constant coefficients divide the $\binom{63}{17}$ anchored root sets. Reciprocity fixes the opposite end of each root polynomial for free. The degree allowance at rate $1/4$ is already enough to contain the remaining middle terms after removing the anchors. The line construction uses a different rate and one pair of reserved coefficients. Its scalar count exceeds the numerical line prescription by more than four million. Both examples transfer to ordinary Reed–Solomon codes by Theorem A.10, with length $n - 2$ and dimension $K - 1$. At these rates the exact transferred rate is $(K - 1)/(n - 2)$, rather than precisely $1/4$ or $1/8$.

6.4 The quadratic field already suffices

The larger scalar count above uses a pole in $\mathbb{F}_{p^4} \setminus \mathbb{F}_{p^2}$. An outside pole is not essential to the counterexample. It is enough to avoid the packet labels μ_M . Choosing the pole inside $\mathbb{F}_{p^2}$ gives fewer available poles, but also keeps the entire construction over the quadratic field.

For example, take $p = 2^{31} - 1$, $n = 2^{26}$, $K = 2^{23}$, and

$$\rho = \frac{1}{8}, \quad \eta = \frac{3}{128} - \frac{2}{n}.$$

Corollary A.5, with $M = 128$, $B = 2^{19}$, $r = 0$, and $h = 19$, gives an $\mathbb{F}_{p^2}$ -valued pair with no correlated agreement at threshold $(\rho + \eta)n$, and more than 2^{57} nearby scalar combinations. Its certified scalar count exceeds $n2^{H_2(\rho)/\eta}$ by more than 400. The radius remains strictly below Elias, and the same example transfers to an ordinary Reed–Solomon code with a prime-field evaluation domain and quadratic alphabet. Appendix D.2 records the exact count and inequalities.

6.5 Shorter codes give larger separations

The number of selected scalar labels depends on the packet group and the reserved coefficients, but not on the number B of coordinates in each packet. Reducing B therefore preserves that number while reducing the length factor in the proposed line threshold. The gap also changes, so the comparison must be recomputed.

Corollary 6.4 (Compact eighth-rate examples). *Let $p = 2^{31} - 1$ and $\rho = 1/8$. Each row below gives a circle code on μ_n , or any rotation, and a line with no correlated agreement at threshold $T = (\rho + \eta)n$. Write J for its certified number of distinct nearby parameters.*

<i>Alphabet</i>	<i>n</i>	<i>K</i>	<i>η</i>	<i>$J/(n2^{H_2(\rho)/\eta})$</i>
$\mathbb{F}_{p^2}$	2048	256	23/1024	$> 2^{22}$
$\mathbb{F}_{p^4}$	8192	1024	79/4096	$> 2^{34}$

Both radii are strictly below the characteristic- p Elias radius. In the first row, the words, witnesses, and labels lie in $\mathbb{F}_{p^2}$; the absence of correlated agreement holds even against the larger circle code over $\mathbb{F}_{p^4}$.

Proof. Use $(M, B, r, h) = (128, 16, 0, 19)$ in Corollary A.5 and $(256, 32, 1, 37)$ in Theorem A.4. The respective agreement counts are 302 and 1182, exceeding the direction-word root bounds $K + B = 272$ and 1056. The label counts are unchanged from the preceding eighth-rate examples. Appendix D.3 gives integer certificates for the two new radii and comparisons. $\square$

7 Implications for hash-based SNARK security

The preceding results concern properties of codes. Their relevance to SNARKs comes from the role those properties play in a soundness proof. This section explains that connection before considering a particular protocol.

7.1 The coding statement used by a reduction

An interactive oracle proof can ask a prover to supply evaluations of polynomials and then test those evaluations through a small number of queries. In a hash-based argument, Merkle commitments bind the prover to the supplied words. The algebraic analysis must still explain what passing the tests says about those words. List decoding and proximity gaps are two ingredients in such analyses [1, 2].

For example, consider a code $C \subseteq E^n$, two fixed words $f_0, f_1 \in E^n$, and a uniformly random challenge $z \in E$. Define

$$B_\theta(f_0, f_1) = \{z \in E : \text{dist}(f_0 + zf_1, C) \leq \theta\}.$$

Suppose a coding theorem guarantees that, unless f_0 and f_1 have correlated agreement with two codewords, the set B_θ has size at most T . Then, for words without that correlated agreement,

$$\Pr_z[\text{dist}(f_0 + zf_1, C) \leq \theta] \leq \min\{1, T/|E|\}.$$

This is a bound on a specified algebraic event. Using it in a soundness proof also requires a reduction from verifier acceptance to that event, with the words fixed before the challenge is sampled. The evaluation field, coefficient field, and challenge field must be the fields covered by the coding theorem.

The size of a decoding list enters a different counting step. If an analysis associates a received word with at most L candidate codewords and proves a failure bound for each candidate, it can apply a union bound over those candidates. A numerical estimate for L is therefore part of the quantitative argument. Replacing the actual list size by a conjectured smaller value can invalidate that argument. It does not determine the probability of the verifier's acceptance event.

7.2 What the counterexamples rule out

Three distinctions are needed to apply the results correctly.

Minimum distance and list size. The root bound proves the usual minimum-distance guarantee for Reed–Solomon codes. It does not bound how many codewords can lie near an arbitrary received word beyond the unique-decoding radius. Theorem 2.2 rules out a uniform $\exp(O_\rho(1/\eta))$ bound over prime fields and evaluation domains, even below the characteristic-based Elias radius. An analysis that needs such a bound must impose further restrictions on the code or use a different quantitative estimate.

A list and the nearby points on a line. A maximum-list bound concerns one received word at a time. It does not say that the nearby codewords for different points on an affine line belong to a common small family. Theorem 5.1 makes this distinction explicit: the global maximum list size is $n - k - 1$, while one affine line has $\binom{n}{k+1}$ nearby points, each with a unique witness. Its gap is $1/n$. This theorem does not settle the analogous question with a separately fixed positive gap.

Uniform estimates and fixed-parameter estimates. A security calculation uses a particular field, domain, rate, and radius. A theorem quantified over all prime-field codes would apply to that code, but a counterexample to the universal theorem need not use that particular code. Conversely, a bound with constants depending on the field or the gap cannot be used uniformly without controlling that dependence. The fixed-gap lift in Theorem 4.4 rules out a prescribed numerical coefficient with a remainder sublinear along its increasing-length family. Its field grows, and its chosen list size remains fixed. These quantifiers are part of the result.

7.3 From coding counterexamples to protocol conclusions

A coding counterexample is enough to refute a coding premise under the same quantifiers. A conclusion about a complete SNARK requires more. Its oracle words may be constrained by an arithmetization, previous commitments, claimed evaluations, and later consistency tests. A soundness analysis must account for those constraints and the distribution and order of all challenges. Passing to a noninteractive protocol through Fiat–Shamir introduces a further cryptographic analysis.

The circle constructions give finite examples in polynomial spaces used by Circle STARKs. The quotient calculations in Appendices B and C examine additional algebraic conditions on derived words. Their statements retain their own sampling, radius, and conditioning hypotheses. In particular, the numerical raw-quotient example lies above the Elias radius, whereas the principal list and affine-line examples lie below it. No full-protocol security level follows from the quotient counts in this paper.

The application below uses this distinction to identify which quantitative premises of a published analysis are contradicted and which questions about the instantiated protocol remain open.

8 Application to S-two’s quantitative coding assumptions

The S-two whitepaper uses list- and line-decodability assumptions in its analysis of polynomial proximity tests [14]. This provides one application of the preceding coding results. We separate three questions: the asserted uniform list bound, the coefficient in a line-decodability threshold, and finite numerical instantiations. They require different counterexamples and different quantifiers.

8.1 The uniform list prescription

Conjecture 1 of the whitepaper takes the form

$$\ell_C(\theta) \leq c_1 2^{c_2 H_2(\rho)/\eta}, \quad \theta = 1 - \rho - \eta, \quad (12)$$

with constants depending only on the rate and a stated range below the characteristic-based Elias radius. Under its universal prime-field and evaluation-domain quantifiers, this prescription is false for every fixed pair of finite positive constants. Theorem 2.2 gives a single sequence with

$$\log_2 \ell_C(\theta) = \Omega_\rho(n), \quad 1/\eta = \Theta_\rho(\sqrt{n \log n}).$$

The logarithm of the right-hand side of (12) is sublinear in n , whereas the constructed list has a logarithm linear in n . Changing fixed constants cannot close that gap.

This conclusion uses growing prime fields and a decreasing gap. It does not assert exponential lists at a fixed positive gap, and it does not rule out stronger guarantees for a restricted class of evaluation domains or a restricted relation between field size and length.

8.2 The quantitative line coefficient and its remainder

Conjecture 2 in Appendix A.5 of the whitepaper has a threshold of the form $a = \ell(\theta)n + o(n)$, with concurrency number n . Substituting the proposed quantitative list bound gives

$$a = c_1 2^{c_2 H_2(\rho)/\eta} n + o(n). \quad (13)$$

Here the asymptotic remainder matters. A finite example, or a sequence whose gap decreases, need not refute a remainder statement made only for each separately fixed gap.

Theorem 4.4 addresses that issue directly. For any fixed constants, it preserves the prescribed rate and chooses a gap fixed along an increasing-length family. A preserved list has size $L > A$, where $A = c_1 2^{c_2 H_2(\rho)/\eta}$. The family has Ln selected nearby witnesses, but no parametrized affine codeword graph contains more than n selected pairs. Its discrepancy from the proposed coefficient is

$$(L - A)n.$$

Since $L - A > 0$ is fixed, no remainder that is $o(n)$ along this family can absorb the discrepancy. The family is selected independently of the remainder. Here $o(n)$ means sublinear along admissible increasing-length families, allowing the prime field to grow; the whitepaper does not explicitly state its field-uniformity convention. The theorem is stated for constants at least one; any smaller positive constants are covered by enlarging them to one.

There are two qualifications. First, the fixed-gap construction has correlated agreement. It refutes the quantitative selected-witness line threshold; it is not a fixed-gap example of global failure of correlated agreement. Second, it does not refute a threshold expressed using the *actual* maximum list size $\ell_C(\theta)$. The constructed selected count is $Ln \leq \ell_C(\theta)n$. Replacing a false numerical estimate for list size and retaining a theorem that uses actual list size are different operations.

Theorem 5.1 addresses the actual-list question separately. At every fixed rate it has $\ell_C(\theta) = n - k - 1$ but exponentially many uniquely nearby line points and maximum concurrency $n - k$. At rate one-half these bounds are $n/2 - 1$ and $n/2$, respectively. Thus even the threshold $\ell_C(\theta)n + o(n)$ fails along this shrinking-gap family. This rules out a familywise interpretation of that threshold; it does not settle a version whose remainder is controlled only after a positive gap has been fixed. The original conjecture does not specify this uniformity convention.

8.3 Finite comparisons on the circle code

The whitepaper’s concrete examples set $c_1 = c_2 = 1$ and omit the asymptotic remainder. At the common parameters of Theorem 6.1, the resulting comparison values are $2^{1/\eta}$ for a list and $n2^{1/\eta}$ for the line threshold. Let L_0 and B_0 denote the list-size and scalar-count lower bounds certified in Appendix A. They give

$$\frac{L_0}{2^{1/\eta}} > 500, \quad \frac{B_0}{n2^{1/\eta}} > 200,000.$$

These are ratios of coding-theoretic counts. They are not multipliers for a complete attack’s success probability and are not security-bit losses. In particular, the logarithm of a list-size ratio is not, by itself, a reduction in a proof system’s security level.

The two bounds also concern the displayed radius. A parameter-table row evaluated at a different gap must be checked at that gap. For example, the value 2^{16} at rate $1/2$ corresponds to $\eta = 1/16$, which differs from $33/1024 - 2/n$. A counterexample to a universal prescription does not automatically supply a counterexample to every individual numerical row.

The circle tensor space and canonical full-coset domains described above are the algebraic spaces appearing in the examined public proving source [15]; Proposition A.12 and Proposition A.13 explain their mathematical identification. This connection makes the fixed-prime examples pertinent to the coding assumptions used for that family of tests.

Concrete parameters and the challenge schedule. The inspected production configuration uses a nominal 96-bit target: 26 bits of query grinding, 70 queries, and blowup factor two. The source’s parameter counter computes $26 + 70 \cdot 1 = 96$, and the project documentation calls its default soundness conjectured [16]. This arithmetic records the parameter convention; a security theorem must justify the contributions attributed to each stage. The whitepaper’s illustrated 100-bit configuration is a different choice [14, Section 5.5].

There is a further distinction between that illustrated calculation and the cited implementation. The former allocates grinding to intermediate challenge stages, while the latter performs query grinding after those challenges and omits the corresponding intermediate checks [17]. Grinding after a challenge cannot be credited to an earlier stage without a supporting argument. A quantitative analysis of the implementation must therefore use its actual challenge schedule and accepted parameters. This issue is separate from the counterexamples to the coding assumptions, and the comparison concerns only the pinned source revisions.

The remaining protocol question. A proximity test inside a proof system does not necessarily receive an arbitrary pair of words. Its inputs can be derived quotient polynomials constrained by earlier commitments, algebraic identities, and sampled evaluation points. Appendix C resolves one of these algebraic conditions: fixed tables and fixed, generally false claimed openings produce raw quotient pencils with a certified joint proximity event. The transfer also handles specified nonlinear batching schedules (Theorem C.4), with explicit quotient-layer realizations. Its numerical raw-quotient example lies above the Elias radius (Section C.4); it is distinct from the below-Elias counterexamples. It does not prove that these inputs satisfy the complete set of commitment, AIR, and FRI conditions. Nor does a count of exceptional scalars determine the cost of finding favorable claims and witnesses or arranging a preceding transcript. Thus the coding results invalidate the stated quantitative justifications under their stated quantifiers; turning them into an attack on a complete protocol is a separate problem.

9 Further coding-theoretic questions

The main open questions concern the restrictions needed for useful coding bounds and their role in concrete soundness analyses.

1. **Restrictions on field size and domain.** What list bounds hold when p is polynomial in n , or when the domain is restricted to a specified multiplicative subgroup or circle coset? The interval construction separates a universal claim from these more structured questions.
2. **The correct dependence on the gap.** For a prescribed relation between p and n , what dependence on η is necessary? The exponential-list family has shrinking gap; the fixed-gap lift retains a fixed list rather than an exponentially growing one.
3. **Bounds for a prescribed code.** Which list and proximity-gap bounds hold for a fixed field and a fixed FFT domain at the radii used in a proof system? A generic-domain existence theorem and a counterexample on an integer interval leave this question open. Such a theorem should expose its finite constants and its dependence on the challenge field.
4. **Finding the large classes.** Can one efficiently describe or enumerate enough of the equal-moment or equal-coefficient classes to match the counting lower bounds? The existence arguments alone do not provide such algorithms.
5. **Lines and actual list size.** What line-decodability statements follow from the actual list size at a fixed positive gap? Theorem 5.1 rules out the proposed linear threshold along a shrinking-gap family, even with unique nearby witnesses on the selected line.

A Circle-code constructions in detail

A.1 Base-field lists for circle evaluation codes

We first construct many codewords close to a single received word. The construction uses the roots of polynomials to choose agreement positions, then fixes a few coefficients so that the remaining polynomial parts have low degree. On the norm-one circle, the leading and trailing coefficients are linked. This lets one impose both sets of conditions at the cost of fixing only one set.

A.1.1 The norm-one circle and Laurent polynomials

Let $p \equiv 3 \pmod{4}$ be prime. Since -1 is not a square in $\mathbb{F}_p$, adjoining a root i of $i^2 = -1$ gives the quadratic extension $F_2 = \mathbb{F}_{p^2}$. Its nontrivial field automorphism is $a \mapsto a^p$, and $i^p = -i$. Thus, for $x, y \in \mathbb{F}_p$,

$$(x + iy)^p = x - iy, \quad (x + iy)^{p+1} = x^2 + y^2.$$

The solutions of $x^2 + y^2 = 1$ can therefore be written as the group

$$U = \{u \in F_2^\times : u^{p+1} = 1\}.$$

This group is cyclic of order $p+1$: it is the kernel of the norm map from $F_2^\times$ to $\mathbb{F}_p^\times$. For $n \mid p+1$, let $\mu_n \subseteq U$ be its subgroup of order n . Multiplying all points by $g \in U$ gives a rotated domain $g\mu_n$.

The coordinate $u = x + iy$ simplifies computations on the circle because $u^{-1} = x - iy$. In particular,

$$x = \frac{u + u^{-1}}{2}, \quad y = \frac{u - u^{-1}}{2i}.$$

A *Laurent polynomial* is a finite sum $\sum_j a_j u^j$ in which negative exponents are allowed. They cause no poles on U , since $u \neq 0$. Its support is the set of exponents with nonzero coefficients. We call a Laurent polynomial with coefficients in F_2 *base-field-real* when

$$a_{-j} = a_j^p \quad \text{for every } j.$$

This coefficient condition makes its values lie in $\mathbb{F}_p$ on U : taking the p th power both conjugates the coefficients and replaces u by u^{-1} . The word “real” here denotes this finite-field condition, not a condition on real numbers.

For even $K = 2s$, the circle polynomial space used below is

$$\mathcal{C}_K = \{A(x) + yB(x) : A, B \in \mathbb{F}_p[x], \quad \deg A < s, \deg B < s\}.$$

It has dimension K as a polynomial-function space on the algebraic circle. Whenever the stated evaluation domain is large enough to make evaluation injective, evaluating these functions gives a dimension- K linear code. We use the term *circle FFT space* for this space; its tensor basis is explained in Proposition A.12. The constructions below lie in its slightly smaller interior Laurent subspace. Working in that subspace makes rotation straightforward.

For a length- n , dimension- K evaluation code, write $\rho = K/n$ for its rate. A word agreeing with a received word in at least $(\rho + \eta)n$ positions lies at relative Hamming distance at most $\theta = 1 - \rho - \eta$. Thus η measures the agreement beyond the rate. Throughout, H_2 denotes binary entropy, and

$$H_p(t) = t \log_p(p-1) - t \log_p t - (1-t) \log_p(1-t).$$

The characteristic- p Elias radius at rate ρ is the solution $r_E(\rho)$ of $H_p(r_E) = 1 - \rho$ on the increasing branch of H_p .

Lemma A.1 (Interior band of the circle FFT space). *Let $p \equiv 3 \pmod{4}$ be prime, let $i^2 = -1$ in $\mathbb{F}_{p^2}$, and write $u = x + iy$ on the norm-one circle. For even $K = 2s$, the dimension- K circle FFT space*

$$\mathcal{C}_K = \mathbb{F}_p[x]_{<s} + y\mathbb{F}_p[x]_{<s}$$

contains every base-field-real Laurent polynomial supported on $[-s+1, s-1]$. This interior subspace is preserved by rotations of the circle.

Proof. For $0 \leq j < s$, the functions $u^j + u^{-j}$ are polynomials in x of degree j , and $(u^j - u^{-j})/i$ are y times polynomials in x of degree $j-1$. The real combinations of these functions span the indicated interior band. Replacing u by a norm-one scalar times u preserves the band and reality. $\square$

Theorem A.2 (Reciprocal-reserve real circle lists). *Let $p \equiv 3 \pmod{4}$ be prime, let $n = MB$ divide $p+1$, and suppose $M \geq 4$ and $B \geq 2$ are powers of two. Choose an even integer d with $0 < d < M$, and an integer $r \geq 0$ such that*

$$h = d + 2r + 2 < M.$$

Set $K = dB$ and $s = K/2$. On $D = \mu_n$, and on any rotation of this domain, there exists an $\mathbb{F}_p$ -valued received word with at least

$$L = \left\lceil \frac{\binom{M-1}{h-1}}{Mp^{2r}} \right\rceil$$

distinct codewords in the exact space $\mathcal{C}_K$ of Lemma A.1, each agreeing with the received word at at least $hB - 2$ positions. The rate and gap are exactly

$$\rho = \frac{d}{M}, \quad \eta = \frac{2r+2}{M} - \frac{2}{n}, \quad \theta = 1 - \rho - \eta.$$

Every witness lies in the strict interior Laurent band, so no boundary-mode identification is needed to transport the construction to a coset.

Proof. For each h -subset $A \subset \mu_M$ containing 1, put

$$F_A(Y) = \prod_{a \in A} (Y - a) = \sum_{j=0}^h f_j Y^j, \quad c = f_0.$$

The integer h is even. Consequently $c \in \mu_M$, and the norm-one roots give the reciprocal coefficient identity

$$f_{h-j} = c f_j^p. \quad (\text{R1})$$

Partition all $\binom{M-1}{h-1}$ anchored subsets by c and the r leading coefficients $f_{h-1}, \dots, f_{h-r}$. There are at most Mp^{2r} classes. In a largest class, containing at least L supports, the bottom r coefficients are fixed as well, by (R1).

Write $T_A(Y) = F_A(Y)/Y^{r+1}$. Let $v(Y)$ consist of its common top $r+1$ and bottom $r+1$ Laurent terms. Thus these are precisely the terms coming from $f_0, \dots, f_r$ and $f_{h-r}, \dots, f_h$. The remainder

$$R_A(Y) = v(Y) - T_A(Y)$$

is an ordinary polynomial of degree at most $d = h - 2r - 2$. Distinct supports give distinct R_A , and the anchor gives $R_A(1) = v(1)$.

For norm-one Y , (R1) implies

$$T_A(Y)^p = c^{-1} Y^{-d} T_A(Y). \quad (\text{R2})$$

The involution in (R2) pairs the retained top and bottom blocks, so v and R_A satisfy the same identity. Choose $\alpha \neq 0$ with $\alpha^p/\alpha = c$, possible because c has norm one. Explicitly, $\alpha = 1 + c^{-1}$ works if $c \neq -1$, and $\alpha = i$ works otherwise. Define

$$V(u) = \alpha u^{-s} v(u^B), \quad S_A(u) = \alpha u^{-s} R_A(u^B).$$

Since $dB = 2s$, (R2) proves that V and S_A are fixed by base-field Frobenius and hence are $\mathbb{F}_p$ -valued on the circle. Both B and s are even, so every S_A has the same value $\beta = \alpha v(1) \in \mathbb{F}_p$ at $u = 1$ and $u = -1$.

Let $y(u) = (u - u^{-1})/(2i)$. Away from these anchors define

$$W(u) = \frac{V(u) - \beta}{y(u)}, \quad P_A(u) = \frac{S_A(u) - \beta}{y(u)}.$$

Give W arbitrary base-field values at the two anchors. For the witness, multiplication of its numerator by u^s gives the polynomial

$$\alpha(R_A(u^B) - v(1)u^s),$$

of degree at most K , divisible by $u^2 - 1$. Therefore P_A is a Laurent polynomial supported on $[-s+1, s-1]$, with removable values at the anchors. Its reality, including at the removable points, follows from the Frobenius identity of the rational expression. Lemma A.1 places it in the exact dimension- K circle space. When K is a power of two, this is the corresponding FFT space.

Off the anchors,

$$W(u) - P_A(u) = \frac{\alpha u^{-s} F_A(u^B)}{u^{B(r+1)} y(u)}.$$

Thus agreement holds exactly on the selected h fibers, except for the two discarded anchors. The map $u \mapsto u^B$ has fibers of size B on D , so there are $hB - 2$ certified agreements. Distinct F_A give distinct P_A ; a nonzero difference in the interior band, after clearing its Laurent monomial, has degree at most $K - 2 < n$, so the codewords are distinct on D . Finally, rotation preserves the real interior band and all agreement counts. $\square$

A.1.2 A numerical example over a Mersenne prime

Take $p = 2^{31} - 1$ and $n = 2^{26}$. For $M = 2048$, $r = 32$, and $h = 1090$, Theorem A.2 gives the exact integer lower bound

$$L = \left\lceil \frac{\binom{2047}{1089}}{2048(2^{31} - 1)^{64}} \right\rceil = 1,199,728,024,606.$$

For comparison, the same theorem with no reserved leading coefficients has $M = 32$, $r = 0$, $h = 18$, and $L = 8,286,954$. Consider the candidate finite list-size bound $2^{H_2(\rho)/\eta}$. At rate $1/2$, its base-two logarithm is $1/\eta$. The following table compares that bound with the two lower bounds.

M	r	h	$\log_2 L$	$1/\eta$	$\log_2 \text{ratio}$
32	0	18	22.98241048	16.00000763	6.98240285
2048	32	1090	40.12584453	31.03033173	9.09551280

The last column is the base-two logarithm of the ratio between the proved list-size lower bound and the candidate upper bound. The pigeonhole argument establishes the existence of the large coefficient class; it does not require enumerating that class.

A.1.3 Strict below-Elias certificate

For the second row, the gap is

$$\eta = \frac{33}{1024} - \frac{2}{2^{26}} = 0.0322265326976776123046875.$$

It is slightly smaller than $1/\log_2 p$, so that convenient sufficient condition must not be used. Direct high-precision evaluation instead gives

$$H_p(1/2 - \eta) = 0.49993479960319703654 \dots < 1/2.$$

The conclusion also has an exact integer/rational certificate. Direct integer arithmetic verifies $p^{100} > 2^{3099}$, hence $\log_2 p > 3099/100$. The binary entropy satisfies $H_2(1/2 - \eta) \leq 1 - 2\eta^2$: its second derivative is at most -4 , and its derivative vanishes at $1/2$. Direct rational arithmetic gives

$$\eta \frac{3099}{100} > 1 - 2\eta^2.$$

Consequently

$$H_p(1/2 - \eta) < (1/2 - \eta) + \frac{H_2(1/2 - \eta)}{\log_2 p} < 1/2.$$

This radius is strictly below characteristic- p Elias. Independently of rounded logarithms, the exact checks $L^8 > 2^{321}$ and $1/\eta < 31031/1000$ certify a list excess greater than 9.094 bits.

A.1.4 A small exact-arithmetic example

An independent implementation over $\mathbb{F}_{31}[i]/(i^2 + 1)$ uses $n = 32$, $M = 16$, $B = 2$, $r = 1$, and $h = 12$. It enumerates all 1,365 anchored supports, finding 1,336 coefficient classes and a largest class of size three. For every member of that class it verifies all reciprocal coefficient identities, real-valued centering, both removable anchors, exact division by $u^2 - 1$, the strict Laurent band $[-7, 7]$, distinctness, and agreement at exactly 22 non-anchor domain positions. This small example checks the algebraic construction; its parameters are not claimed to violate the numerical candidate bound.

The finite-field checks use exact field arithmetic and integers; the displayed logarithms use 100-digit decimal arithmetic. Negative controls with $(M, r) = (512, 7)$ and $(1024, 15)$ have larger counts but are above Elias for this list construction and therefore do not give examples below Elias for this list construction.

A.2 Affine families from simultaneous coefficient constraints

We now seek a different phenomenon: many scalar combinations of two received words have nearby codewords, even though the two original words have no nearby codewords on a sufficiently large common set of positions. Such a scalar-parameterized affine family is also called a *pencil*.

Throughout this section, $p \equiv 3 \pmod{4}$ is prime, $F_2 = \mathbb{F}_{p^2} = \mathbb{F}_p(i)$ with $i^2 = -1$, and $E = \mathbb{F}_{p^4}$. A bar denotes the nontrivial automorphism of $F_2/\mathbb{F}_p$. The norm-one group

$$U = \{u \in F_2^\times : u\bar{u} = 1\}$$

has order $p + 1$. For each $n \mid p + 1$, write μ_n for its unique subgroup of order n . We use the split coordinate $u = x + iy$ on the circle; then $x = (u + u^{-1})/2$ and $y = (u - u^{-1})/(2i)$.

For $m \geq 1$, define the Laurent spaces

$$\mathcal{L}_m^- = \text{span}_E\{u^j : -(m-1) \leq j \leq m-1\}, \quad \mathcal{L}_m^+ = \text{span}_E\{u^j : -m \leq j \leq m\}.$$

If $D \subseteq U$, evaluation of either space on D will be denoted by the same symbol when the domain is unambiguous. We consider any code $C \subseteq E^D$ satisfying

$$\mathcal{L}_m^- \subseteq C \subseteq \mathcal{L}_m^+. \tag{14}$$

The dimension- $2m$ circle FFT space satisfies this condition, as shown in Section A.4. The constructions below place every witness in the smaller space, and rule out correlated agreement even in the larger one.

For words $f, g \in E^D$, let $\text{agr}_D(f, g)$ be the number of coordinates on which they agree. A pair (f_0, f_1) has correlated agreement at threshold T with C if there exist $P_0, P_1 \in C$ agreeing with the two words simultaneously on at least T coordinates. When no such pair exists, a scalar $z \in E$ is *globally bad at threshold T* if

$$\text{agr}_D(f_0 + zf_1, P_z) \geq T$$

for some $P_z \in C$. A *bank* is a set of distinct such scalars, together with one witness for each scalar. Here the bank records only which scalars have nearby codewords; it does not require those codewords to depend affinely on the scalar.

Lemma A.3 (Global failure and line-decodability). *Suppose (f_0, f_1) has no correlated agreement at threshold T on a domain of length n . Select a witness P_z with at least T agreements for each scalar in a bank. No affine line of codewords contains more than $n - T + 1$ of the selected pairs (z, P_z) . In particular, it contains at most n such pairs.*

Proof. Suppose $P_z = A + zB$ for q distinct selected scalars. Let a be the number of coordinates at which both $f_0 = A$ and $f_1 = B$. At every other coordinate, the affine equality $f_0 + zf_1 = A + zB$ can hold for at most one scalar. Hence

$$qT \leq qa + n - a.$$

Since $a \leq T - 1$, this gives $q(T - a) \leq n - a$, and therefore $q \leq n - T + 1$. $\square$

Consequently, a bank larger than a proposed line-decodability threshold contradicts the definition with concurrency number n . This observation will connect the explicit pencils to the finite threshold discussed below.

Theorem A.4 (Simultaneous coefficient reserve pencil). *Let $n = MB \mid p + 1$, where $B \geq 2$ is even. Let $r \geq 0$ and $2r + 4 \leq h \leq M - 1$, and put*

$$a = 2r + 3, \quad K = (h - a)B = 2m, \quad s = m - 1, \quad T = hB - 2.$$

Assume $K + B < T \leq n$, and let C be any code on $D = \mu_n$ satisfying (14). Define

$$N = \left\lceil \frac{\binom{M-1}{h-1}}{Mp^{2r}} \right\rceil, \quad Q = p^4 - p^2, \quad d = h - 2r - 2. \quad (15)$$

There exist words $\gamma_0, \gamma_1 \in E^D$ which have no correlated agreement at threshold T with C , and which have a bank of size at least

$$\left\lceil \frac{NQ}{Q + (d - 1)(N - 1)} \right\rceil. \quad (16)$$

Each selected witness lies in $\mathcal{L}_m^-$, and agrees on at least T coordinates outside $\{1, -1\}$. The same conclusion holds on any rotated domain $g\mu_n \subseteq U$.

Proof. Put $G = \mu_M$. For every h -element subset $A \subseteq G$ containing 1, write

$$F_A(Y) = \prod_{a \in A} (Y - a) = \sum_{j=0}^h f_{A,j} Y^j, \quad f_{A,h} = 1.$$

Partition these subsets by $c = f_{A,0}$ and the r coefficients $f_{A,h-1}, \dots, f_{A,h-r}$. The constant coefficient has at most M possible values, and each other coefficient lies in F_2 , so there are at most Mp^{2r} classes. Fix a class of size at least N , and retain exactly N distinct subsets in it.

The norm-one condition on the roots gives the identity

$$F_A(Y) = cY^h \overline{F_A}(1/Y), \quad f_{A,j} = c \overline{f_{A,h-j}}. \quad (17)$$

Thus the bottom r coefficients are common to the class as well. In $F_A(Y)/Y^{r+1}$, let $v(Y)$ be the sum of the common terms with indices $0 \leq j \leq r$ and $h - r \leq j \leq h$. Then

$$R_A(Y) := v(Y) - F_A(Y)/Y^{r+1}$$

is an ordinary polynomial of degree at most $d = h - 2r - 2$. The polynomials R_A are distinct, and all satisfy $R_A(1) = v(1)$.

Choose a pole $\tau \in E \setminus F_2$, to be specified below. Define

$$f_0(u) = \frac{v(u^B)}{u^B - \tau}, \quad f_1(u) = \frac{1}{u^B - \tau}, \quad \gamma_j(u) = u^{-s} \frac{f_j(u) - f_j(1)}{u^2 - 1}.$$

At $u = \pm 1$ use the removable values. There are no other denominator zeros on D : every u^B lies in F_2 , while $\tau \notin F_2$. For each A , put

$$z_A = -R_A(\tau), \quad Q_A(Y) = \frac{R_A(Y) - R_A(\tau)}{Y - \tau}, \quad P_A(u) = u^{-s} \frac{Q_A(u^B) - Q_A(1)}{u^2 - 1}.$$

The polynomial Q_A has degree at most $d - 1 = h - a = K/B$. Since B is even, its composed numerator vanishes at $u = \pm 1$. Dividing by $u^2 - 1$ leaves degree at most $K - 2$, and multiplying by u^{-s} places P_A in the Laurent band $[-s, s]$. Thus $P_A \in \mathcal{L}_m^- \subseteq C$. At every nonanchor point with $u^B \in A$, the identity $v(u^B) = R_A(u^B)$ holds; it also holds at the subtracted anchor value. Therefore $\gamma_0(u) + z_A \gamma_1(u) = P_A(u)$ at at least $hB - 2 = T$ coordinates outside $\{1, -1\}$.

To rule out correlated agreement, it suffices to rule out sufficient agreement with γ_1 alone. Put

$$S_B(u) = 1 + u^2 + \dots + u^{B-2}.$$

Cancellation yields

$$\gamma_1(u) = -\frac{u^{-s} S_B(u)}{(1 - \tau)(u^B - \tau)}. \quad (18)$$

For $P \in \mathcal{L}_m^+$, the function $H(u) = u^m P(u)$ is an ordinary polynomial of degree at most K . Every agreement with γ_1 is a root of

$$(1 - \tau)(u^B - \tau)H(u) + u S_B(u). \quad (19)$$

This polynomial is nonzero: the second summand has degree $B - 1$, whereas the first has degree at least B when $H \neq 0$. Its degree is at most $K + B$. Hence no word in the entire outer-band code agrees with γ_1 on more than $K + B < T$ coordinates. Formula (18) also holds at the removable anchor values.

It remains to select a pole with sufficiently many distinct labels. For $A \neq A'$, the nonzero polynomial $R_A - R_{A'}$ has degree at most d and has the root $1 \in F_2$. It therefore has at most $d - 1$ roots outside F_2 . Averaging over the Q possible poles, some pole has at most $(d - 1)N(N - 1)/(2Q)$ colliding unordered pairs. If the multiplicities of its distinct labels are $b_1, \dots, b_t$, then

$$\sum b_i = N, \quad \sum b_i^2 \leq N + (d - 1)N(N - 1)/Q.$$

Cauchy–Schwarz gives $t \geq NQ/[Q + (d - 1)(N - 1)]$, proving (16). Finally, substituting u/g transports all words to $g\mu_n$. Rotation preserves both Laurent bands and all agreement counts. $\square$

Corollary A.5 (Poles in the quadratic coefficient field). *Under the hypotheses of Theorem A.4, the defining words, selected witnesses, and scalar labels can all lie in $F_2 = \mathbb{F}_{p^2}$. With*

$$Q_2 = p^2 - M,$$

the guaranteed number of distinct labels is

$$\left\lceil \frac{NQ_2}{Q_2 + (d - 1)(N - 1)} \right\rceil.$$

The pair has no correlated agreement at threshold T , even against outer-band codewords over an extension of F_2 .

Proof. Choose the pole from $F_2 \setminus \mu_M$. The denominators only require $u^B \neq \tau$ and $1 \neq \tau$, so every pole in this set is allowed, including zero. All coefficients and values in the pencil formulas then belong to F_2 . A difference $R_A - R_{A'}$ still has the root 1, which is excluded from the pole set, and therefore has at most $d - 1$ roots among its Q_2 elements. The same collision average proves the count with Q_2 in place of Q . The nonzero polynomial in (19) has degree at most $K + B < T$ over any ambient extension. Its root bound, and hence the absence of correlated agreement, are unchanged. $\square$

Remark A.6 (Existence and explicitness). The formulas specify a concrete family of pencils. The large coefficient class and a sufficiently good pole are guaranteed by counting; a large coefficient class or a suitable pole need not have been enumerated to prove the lower bound. We distinguish this existence-certified bank from a bank whose labels and witnesses have all been listed explicitly.

A.2.1 A numerical affine-family example over a Mersenne prime

Let $p = 2^{31} - 1$ and take

$$M = 1024, \quad r = 15, \quad h = 545, \quad n = 2^{26}, \quad B = 2^{16}.$$

Then $a = 33$, $K = n/2$, $d = 513$, and

$$T = 545B - 2, \quad \theta = 1 - T/n = \frac{479}{1024} + \frac{2}{n}, \quad \eta = \frac{33}{1024} - \frac{2}{n}.$$

The theorem gives the integer bank lower bound

$$30,608,283,697,878,843,892,998, \tag{20}$$

whose base-two logarithm exceeds 74.6963402382.

At exact rate $1/2$, the finite prescription $n2^{H_2(1/2)/\eta}$ has logarithm less than 57.0303317264. The excess in the logarithm is greater than 17.6660085118 bits.

For clarity, define the characteristic- p Elias radius $r_E(\rho)$ by $H_p(r_E) = 1 - \rho$, with

$$H_p(t) = t \log_p(p - 1) - t \log_p t - (1 - t) \log_p(1 - t).$$

The above distance is strictly below $r_E(1/2)$. Equivalently, the quantity

$$(1/2 - \theta) \log_2 p - H_2(\theta) - \theta \log_2(1 - 1/p)$$

is positive; its value exceeds 0.0020212123. The exact count and high-precision entropy comparison are reproduced in the accompanying `counterexample_parameters.py` and its machine-readable output. The probability that a uniform scalar in E belongs to the certified bank is at least the count in (20) divided by p^4 . This probability describes the scalar-combination event for the two words in the theorem.

Remark A.7 (The comparison radius matters). This example contradicts a universal finite threshold prescription at the displayed radius. It does not, by itself, contradict each numerical row of a parameter table that evaluates the prescription at a different radius. For example, substituting $\eta = 1/16$ into $2^{1/\eta}$ gives 2^{16} ; that evaluates the candidate at a different radius from the one used here.

A.2.2 A sharper bound on selected-witness concurrency

The single-word agreement bound in the reserve construction gives a much stronger concurrency conclusion than the generic bound in Lemma A.3.

Lemma A.8 (Incidence bound from a quantitative agreement deficit). *Let $C \subseteq E^D$ be a linear code on n coordinates, and let $0 \leq a_0 < T \leq n$ be integers. Suppose a word f_1 agrees with every codeword of C in at most a_0 coordinates. For any word f_0 and any selection of codewords p_z , indexed by distinct scalars, with $\text{agr}_D(f_0 + z f_1, p_z) \geq T$, every parametrized affine codeword graph contains at most*

$$\left\lfloor \frac{n - a_0}{T - a_0} \right\rfloor$$

of the selected pairs (z, p_z) .

Proof. Fix a graph $P_0 + z P_1$ containing q selected pairs, and let a count the coordinates at which $f_0 = P_0$ and $f_1 = P_1$ simultaneously. Then $a \leq a_0$. At every other coordinate the affine equality holds for at most one of the distinct selected scalars. Counting agreements gives $qT \leq qa + n - a$, so

$$q \leq \frac{n - a}{T - a} = 1 + \frac{n - T}{T - a} \leq \frac{n - a_0}{T - a_0}.$$

The final inequality uses $n \geq T$ and $a \leq a_0 < T$. Taking the integer part proves the claim. $\square$

Corollary A.9 (At most fifteen selected witnesses on one graph). *Every parametrized affine codeword graph contains at most 15 selected pairs from the M31 bank in (20). The same upper bound holds after the two-anchor puncturing and stereographic scaling of Theorem A.10.*

Proof. The root certificate (19) bounds agreement of γ_1 with every outer-band codeword by $a_0 = K + B$. In the displayed M31 instance, $n = 1024B$, $K = 512B$, $T = 545B - 2$, and $B = 65536$. Lemma A.8 therefore gives

$$q \leq \left\lfloor \frac{511B}{32B - 2} \right\rfloor = 15.$$

Indeed, subtracting $15(32B - 2)$ from the numerator gives $31B + 30 > 0$, whereas $16(32B - 2) - 511B = B - 32 > 0$.

Puncturing removes only coordinates outside the certified T agreements, and cannot increase the maximum agreement with γ_1 beyond a_0 . The nonzero stereographic scaling preserves equalities; every transformed codeword lifts to the interior band, which is contained in the outer band. Thus the transformed code of length $n - 2$ satisfies

$$q \leq \left\lfloor \frac{511B - 2}{32B - 2} \right\rfloor = 15$$

by the same lemma. This is an upper bound on selected-witness concurrency, not a claim that a graph attaining fifteen witnesses has been exhibited. $\square$

A.3 Transfer to prime-domain Reed–Solomon codes

Theorem A.10 (Two-anchor puncturing and stereographic scaling). *Let $p \equiv 3 \pmod{4}$, let E contain $\mathbb{F}_{p^2}$, and let $D = \mu_n$, where $n \mid p + 1$ is even and $K = 2m < n$. Suppose words γ_0, γ_1 and a bank of distinct scalars have witnesses in $\mathcal{L}_m^-$, each with at least T agreements outside $\{1, -1\}$. Suppose the pair has no correlated agreement at threshold T with $\mathcal{L}_m^-$. Then there is a set $D' \subseteq \mathbb{F}_p$*

of size $n - 2$ and a pair of words over E with the same bank and the same agreement threshold for the ordinary Reed–Solomon code

$$\text{RS}_{<K-1}(E, D').$$

This code has exact rate $(K - 1)/(n - 2)$, and the transformed pair has no correlated agreement at threshold T .

Proof. Delete the two anchors. Writing $u = x + iy$, map each retained point to

$$t = \frac{y}{1 + x} \in \mathbb{F}_p.$$

Stereographic parametrization is injective away from $(-1, 0)$, which was deleted, and has inverse

$$x = \frac{1 - t^2}{1 + t^2}, \quad y = \frac{2t}{1 + t^2}, \quad u = \frac{1 + it}{1 - it}.$$

The resulting set D' has $n - 2$ distinct finite points. Since $p \equiv 3 \pmod{4}$, the function $w(t) = (1 + t^2)^{m-1}$ is nonzero on $\mathbb{F}_p$. For every $j \in [-(m - 1), m - 1]$,

$$w(t)u^j = (1 + it)^{m-1+j}(1 - it)^{m-1-j}. \quad (21)$$

These are polynomials of degree at most $2m - 2 = K - 2$, and they form a basis of that polynomial space. Indeed they are the images of the homogeneous monomial basis of binary forms of degree $2m - 2$ under the invertible change of variables $(X, Y) \mapsto (X + iY, X - iY)$. This proof is valid in positive characteristic and does not require any binomial coefficient to be nonzero.

Multiply each retained coordinate of each word and witness by $w(t)$. Equation (21) shows that the witness code becomes exactly $\text{RS}_{<K-1}(E, D')$. All equality tests and scalar labels are preserved by nonzero coordinate scaling, so every witness keeps its T guaranteed agreements. Any correlated agreement of the transformed pair would lift to two elements of $\mathcal{L}_m^-$ agreeing on the same original coordinates, a contradiction. Finally,

$$\frac{\dim \text{RS}_{<K-1}(E, D')}{|D'|} = \frac{K - 1}{n - 2}.$$

This equals $1/2$ when $K = n/2$. □

Corollary A.11 (Transfer of the reserve bank). *In Theorem A.4, suppose $h = M/2 + a$, where $a = 2r + 3$, so that $K = n/2$. The same bank exists for an ordinary Reed–Solomon code with a prime-field domain, extension alphabet E , exact rate $1/2$, length $n' = n - 2$, and gap*

$$\eta' = \frac{T}{n - 2} - \frac{1}{2} = \frac{aB - 1}{n - 2}.$$

Together with Lemma A.3, a bank larger than $(n - 2)2^{1/\eta'}$ contradicts the corresponding finite line-decodability threshold with concurrency number $n - 2$.

The M31 instance above satisfies this inequality as well. It is therefore an example under the literal prime-field-domain and extension-alphabet quantifiers of the finite prescription, rather than only a circle-code analogue. The transfer uses a change of coordinates and nonzero coordinate scaling. It therefore preserves equality sets and scalar labels exactly.

The accompanying exact-arithmetic fixture checks all 189 archived circle witnesses over $\mathbb{F}_{81914}$ after this transformation. They become degree-at-most-14 polynomials on 30 distinct prime-field coordinates, retain 189 distinct labels, and each has at least 20 agreements. The no-correlated-agreement argument uses the algebraic bound $18 < 20$; it is not represented as an exhaustive enumeration of all codewords.

A.4 Circle polynomial spaces and coset domains

For completeness, we verify two algebraic facts used by the constructions: the relation between the circle tensor basis and Laurent bands, and the relation between a full coset and its two half-cosets. Neither fact depends on an ordering of evaluation coordinates.

Proposition A.12 (The circle tensor basis). *For $K = 2m$ a power of two, the circle FFT tensor basis spans*

$$C_K = \{A(x) + yB(x) : \deg A < m, \deg B < m\}.$$

After scalar extension to E it satisfies $\mathcal{L}_m^- \subseteq C_K \subseteq \mathcal{L}_m^+$.

Proof. The tensor factors are $y, x, \pi(x), \pi^2(x), \dots$, with $\pi(x) = 2x^2 - 1$. Products not using y have distinct leading degrees $0, \dots, m-1$ and nonzero leading coefficients, so they form a basis of the polynomials in x of degree less than m . Multiplication by y gives the second summand. Substitution of $x = (u + u^{-1})/2$ and $y = (u - u^{-1})/(2i)$ proves the outer-band containment. The functions $1, x, \dots, x^{m-1}, y, xy, \dots, x^{m-2}y$ span the inner band. The same basis remains a basis after extending coefficients from $\mathbb{F}_p$ to E . $\square$

Proposition A.13 (Canonical domains). *Suppose n is even and $2n \mid p+1$. If $g \in U$ has order $2n$, the canonical coset $g\mu_n$ is the disjoint union*

$$\{g^{1+4j} : 0 \leq j < n/2\} \cup \{g^{-(1+4j)} : 0 \leq j < n/2\}.$$

Thus describing a canonical domain as two conjugate half-cosets changes only the ordering of its points.

Proof. In additive point-index notation modulo $2n$, the full odd coset is $\{1 + 2j : 0 \leq j < n\}$. Its half-coset representation is $\{1 + 4j : 0 \leq j < n/2\}$ and its negatives. These two half-cosets are disjoint and their union is exactly the full odd coset. Passing to the split multiplicative coordinate gives $g\mu_n$. $\square$

Consequently, a construction on a full multiplicative coset also applies when its coordinates are listed using the two half-cosets. Rotation preserves both Laurent bands.

Remark A.14 (Sizes and target spaces). The M31 circle has order $p+1 = 2^{31}$; a canonical coset uses a subgroup of twice its own size and has maximum length 2^{30} . Here n denotes evaluation length. For example, the rate-1/2 code at length 2^{26} has dimension 2^{25} . The full outer-band circle code has dimension $K+1$, rather than K ; the proof excluding correlated agreement still covers it, but its rate is $1/2 + 1/n$ and any comparison naming that code must use that rate.

A.5 What the finite counterexamples establish

The list construction and the affine-family construction address two different finite bounds. At rate $1/2$, the first gives a list larger than $2^{1/\eta}$. The second gives more than $n2^{1/\eta}$ exceptional scalars, with no correlated agreement at the prescribed threshold. The Reed–Solomon transfer shows that the latter phenomenon also occurs with a prime-field evaluation domain and an extension-field alphabet.

One must distinguish such finite statements from a proposed asymptotic threshold of the form

$$a = \ell(\theta)n + o(n), \quad \ell(\theta) \leq c_1 2^{c_2 H_2(\rho)/\eta}, \quad \theta = 1 - \rho - \eta.$$

The displayed numerical examples contradict the remainder-free choice $c_1 = c_2 = 1$ at their stated fields, lengths, and radii. A single finite example does not determine the unspecified quantifiers in $o(n)$ and does not refute every choice of constants c_1, c_2 .

For an asymptotic claim, the field, domain length, and gap must be allowed to vary in a stated way. In particular, an ordinary Reed–Solomon code with a prime-field evaluation domain has length at most p . Holding p fixed while allowing an arbitrary prime-dependent remainder leaves no unbounded sequence of such lengths. The separate fixed-gap lifting construction in this manuscript gives an increasing-length family for the quantitative coefficient: its discrepancy is linear and cannot be absorbed by a sublinear remainder for that family. That selected-graph construction has correlated agreement and is distinct from the globally non-correlated affine families constructed here. Neither argument refutes a line bound whose coefficient validly bounds the actual list size.

A.6 The common-core form of the circle pencil

The anchored construction in Theorem A.4 can be written in the fixed-core form familiar from the Proximity Prize’s orbit pencil [13]. This is a change of presentation, with the same witnesses and agreement counts.

Use the theorem’s notation, and fix one support A_0 . For each support $A = \{1\} \cup U$, write

$$F_A(Y) = (Y - 1)G_U(Y), \quad D_U(Y) = \frac{G_{U_0}(Y) - G_U(Y)}{Y^{r+1}}.$$

The common leading and trailing coefficients imply $\deg D_U \leq h - 2r - 3 = K/B$. Since $R_A - R_{A_0} = (Y - 1)D_U$, the old scalar labels and the new labels $t_U = D_U(\tau)$ satisfy

$$z_A - z_{A_0} = (1 - \tau)t_U.$$

Set $Q_U = (D_U - t_U)/(Y - \tau)$ and $S_B(u) = (u^B - 1)/(u^2 - 1)$. Then the new pencil and witnesses are

$$g_0 = \frac{u^{-s}S_B(u)G_{U_0}(u^B)}{u^{B(r+1)}(u^B - \tau)}, \quad g_1 = -\frac{u^{-s}S_B(u)}{u^B - \tau}, \quad W_U = u^{-s}S_B(u)Q_U(u^B).$$

Their difference is

$$g_0 + t_U g_1 - W_U = \frac{u^{-s}S_B(u)G_U(u^B)}{u^{B(r+1)}(u^B - \tau)}.$$

The zeros consist of the $h - 1$ varying fibers and the fixed core of $B - 2$ points with $u^B = 1$ and $u \neq \pm 1$. The agreement count is $(h - 1)B + B - 2 = hB - 2$. Also $\deg(S_B Q_U(u^B)) \leq K - 2$, so the centered witnesses lie in the required interior Laurent band.

Direct subtraction in the original formulas gives

$$g_0 = \gamma_0 + z_{A_0}\gamma_1 - P_{A_0}, \quad g_1 = (1 - \tau)\gamma_1, \quad W_U = P_A - P_{A_0}.$$

Thus an invertible affine change of scalar and a codeword translation identify the two presentations. The number of labels and the failure of correlated agreement are unchanged. In particular, introducing this common-core notation does not itself improve the existing numerical bound.

B A two-fiber pencil in raw quotient form

This section addresses one algebraic compatibility condition with the conjugate-line quotient, using the unweighted formula in² the source. It does not construct an accepted Stwo transcript. In

²Source checked at Stwo revision 826591c6c371376810ca8213b5812d5daf6d5092, `core/constraints.rs` and `core/pcs/quotients.rs`.

particular, the claimed openings below are zero and generally false.

Write $p = 2^{31} - 1$, $H = \mathbb{F}_p[i]$, $E = H[w]$, where $i^2 = -1$ and $w^2 = \kappa = 2 + i$. Conjugation over H sends w to $-w$. Let $n = 2M \mid p + 1$, $K = 2m \mid p + 1$, with m even and $m + 3 \leq M - 1$. The circle code C_K contains every Laurent polynomial with exponents between $1 - m$ and $m - 1$, and is contained in the band $[-m, m]$.

Proposition B.1 (Fixed tables realizing a two-fiber pencil). *There are two $\mathbb{F}_p$ -valued tables on μ_n , fixed independently of the sample, and an out-of-domain circle sample over E , such that their normalized raw quotients r_0, r_1 with claimed values zero have at least*

$$J = \left\lceil \frac{NQ}{Q + m(N - 1)} \right\rceil, \quad N = \left\lceil \frac{\binom{M-1}{m+2}}{M} \right\rceil, \quad Q = \frac{p^2 - 1}{2},$$

distinct parameters α for which $r_0 + \alpha r_1$ agrees with C_K on at least $K + 4$ coordinates. The direction r_1 agrees with any codeword on at most $K + 2$ coordinates, so the pencil has no common agreement on $K + 4$ coordinates.

Proof. Choose a class of $(m + 2)$ -subsets $U \subset \mu_M \setminus \{1\}$ with common constant coefficient c in $G_U(Y) = \prod_{a \in U} (Y - a)$. There are at least N members. Fix U_0 , and set

$$D_U = (G_{U_0} - G_U)/Y, \quad \deg D_U \leq m.$$

These polynomials are distinct. Average collisions of their evaluations over the Q nonsquares $\tau \in H^*$. Every difference has at most m roots, so Cauchy–Schwarz gives a nonsquare τ with at least J distinct labels $t_U = D_U(\tau)$. Every element of μ_M is a square in H , so these poles avoid the domain.

Put $Y = u^2$ and

$$g_0 = \frac{u^{1-m} G_{U_0}(u^2)}{u^2(u^2 - \tau)}, \quad g_1 = -\frac{u^{1-m}}{u^2 - \tau}, \quad W_U = u^{1-m} \frac{D_U(u^2) - t_U}{u^2 - \tau}.$$

The last expression lies in the strict Laurent band. The difference $g_0 + t_U g_1 - W_U$ is $u^{1-m} G_U(u^2)/[u^2(u^2 - \tau)]$, giving $2(m + 2)$ agreements. For any codeword C , multiplying $g_1 - C$ by $u^m(u^2 - \tau)$ gives a nonzero polynomial of degree at most $K + 2$. Nonzeroness follows by evaluating at a root of $u^2 - \tau$, where the remaining term is $-u$. This proves the direction bound.

Choose $a_0 \neq 0$ with $a_0^p/a_0 = c$: take $a_0 = 1 + c^{-1}$ if $c \neq -1$, and $a_0 = i$ otherwise. The two fixed tables are

$$f_0(u) = a_0 u^{-(m+2)} G_{U_0}(u^2), \quad f_1(u) = u^m + u^{-m}.$$

Conjugating coefficients and using $u^p = u^{-1}$ proves that both are $\mathbb{F}_p$ -valued. Neither depends on τ .

Because κ and τ are nonsquares in H , choose $b \in H^*$ with $\kappa b^2 = \tau$. Set $a = wb$ and

$$z_x = (a + a^{-1})/2, \quad z_y = (a - a^{-1})/(2i).$$

These coordinates satisfy the circle equation and are anti-invariant under conjugation over H . Writing $z_x = wX_1, z_y = wY_1$, we have $Y_1 \neq 0$ and

$$U_z = (x - (X_1/Y_1)y)/2 = L_0 u^{-1}(u^2 - \tau), \quad L_0 = \frac{1}{2(1 - \tau)}.$$

For claimed opening zero, the raw conjugate-line formula is $q = (\bar{z}_y - z_y)f/[(-x)Y_1 - (-y)X_1]$, hence $q/w = f/U_z$. Thus, with $r_j = q_j/w$, direct multiplication gives

$$r_0 = \lambda_0 g_0, \quad r_1 = \lambda_1 g_1 + P_\tau,$$

where

$$\lambda_0 = a_0/L_0, \quad \lambda_1 = -(\tau^m + 1)/L_0, \quad P_\tau = \frac{u^{1-m}}{L_0} \sum_{j=0}^{m-1} \tau^{m-1-j} u^{2j} \in C_K.$$

Here $\lambda_1 \neq 0$: otherwise $\tau^{2m} = 1$, which would make τ a square in H , since $2m \mid p+1$. For $\alpha_U = \lambda_0 t_U / \lambda_1$, the witness is $\lambda_0 W_U + \alpha_U P_\tau$. The scalar changes and codeword shift preserve distinct labels, agreement counts, and the direction bound. Multiplication by w also preserves membership in the E -linear code. $\square$

Finite instance and remaining protocol gap. Taking $(n, K, M, m) = (128, 32, 64, 16)$ gives agreement 36/128 and $N = 40,448,653,414,754$, yielding $J = 40,437,303,920,543$. The expression above is an exact lower bound; no enumeration of this bank is claimed. A separate deterministic fixture over $p = 127$, with the same extension polynomial, checks the raw quotient identities and agreement counts for multiple nonsquare poles, holding the two base-field tables fixed throughout. An additional fixture over the actual M31 field checks three supports at one nonsquare pole: all 1,536 identities pass, with 36 agreements for each support. This result concerns a specially chosen sample family. It does not establish its probability under Stwo’s sampler, map the actual challenge power schedule or its precise evaluation cosets, or address interpolation and extension of the raw quotient subdomain. Nor does it show that the tables and false openings survive the remaining proof checks. The bounds for larger fibers do not automatically apply to this two-fiber case.

There is a concrete reason to keep the sampler issue separate. Under an idealized uniform sample from the E -rational circle and an independent uniform batching parameter in E , this particular family is rare. The map $a \mapsto ((a + a^{-1})/2, (a - a^{-1})/(2i))$ bijects E^* with that circle. Samples used here have $a \in wH^*$, a fraction $1/(p^2 + 1)$ of the circle. All constructed batching labels lie in H , a fraction $1/p^2$ of E . Thus the event consisting of this sample family and these labels has probability at most $1/[p^2(p^2 + 1)]$, approximately 2^{-124} , even before limiting the label bank. This is an upper bound for the constructed event in the stated idealized model, not a soundness bound for Stwo. In particular, raw quotient compatibility by itself does not establish a practical attack or a reduction below a 96-bit security target.

C From large real lists to raw quotient pencils at typical samples

Sampling outside the evaluation domain to constrain candidate codewords is the established DEEP technique [3]. Here we study large explicit families under the conjugate-line quotient, fixed claims, and prescribed scalar schedules.

The special sample restriction in Appendix B can be removed. We first prove a bound with sample-dependent claimed openings; Theorem C.2 then fixes the claims before the sample and improves the joint probability bound. Sample-dependent claims are consistent with the inspected source order: column commitments precede the sample, while claimed values are mixed before the batching challenge. Both results concern raw quotient agreement; neither proves full proof acceptance.

Theorem C.1 (A list-to-quotient transfer). *Let $p = 2^{31} - 1$, $H = \mathbb{F}_{p^2}$ and $E = \mathbb{F}_{p^4}$, with the tower and normalization of Appendix B. Let $\mathcal{D}$ be a set of n base-field circle points and $K = 2m$. Suppose a fixed base-field word f has N distinct real Laurent polynomials F_j , supported in $[-m, m]$, each agreeing with f on at least $T > K + 2$ points of $\mathcal{D}$. Set*

$$S = (p^2 - 1)^2, \quad D_* = \left\lceil \frac{NS}{S + 2K(N - 1)} \right\rceil, \quad J = \left\lceil \frac{D_*}{p^2} \right\rceil.$$

For at least half of the S admissible circle samples described below, there is a claimed value v such that the two fixed tables $(f, 0)$, with claimed values $(v, 1)$, have normalized raw quotients r_0, r_1 for which at least J distinct $\alpha \in H$ satisfy

$$r_0 + \alpha r_1 \text{ agrees with } C_K(H) \text{ on at least } T \text{ coordinates.}$$

The direction r_1 has at most $K + 2$ agreements with any codeword, so these pencils have no common agreement on T coordinates. The claims are chosen after the sample and before α .

Proof. Parameterize the E -circle by $z_x = (a + a^{-1})/2$, $z_y = (a - a^{-1})/(2i)$, $a \in E^*$. Call a admissible when $a \notin H$ and $z_y \notin H$. For $a \notin H$, the second condition fails exactly when $a\bar{a} = -1$. The sets H^* and this norm fiber have sizes $p^2 - 1$ and $p^2 + 1$, and intersect at $\pm i$. Their complement therefore has size S .

For $j \neq k$, multiplying $F_j - F_k$ by u^m gives a nonzero polynomial of degree at most K . Thus at most K admissible values of a cause $F_j(z) = F_k(z)$. The expected number of unordered collisions is at most $\binom{N}{2}K/S$. By Markov's inequality, at least half the admissible samples have at most $N(N-1)K/S$ collisions. Cauchy-Schwarz then gives at least D_* distinct values among $F_j(z)$. One of the p^2 additive H -cosets in E contains at least J of them. Write this coset as $v + H$, and retain one index for each distinct value. The labels $\alpha_j = F_j(z) - v$ are distinct and belong to H .

Use the source functions U_z and ℓ_b defined in Appendix B and the preceding note. Explicitly, for $z_x = X_0 + wX_1$, $z_y = Y_0 + wY_1$, and $b = B_0 + wB_1$,

$$U_z = (x - X_0)/2 - (X_1/Y_1)(y - Y_0)/2, \quad \ell_b = B_0 + (B_1/Y_1)(y - Y_0).$$

The polynomial uU_z has degree two and distinct nonzero roots $a, \bar{a}$. The numerator $F_j - \ell_{F_j(z)}$ vanishes at both roots. Indeed $\ell_b(z) = b$, and conjugating over H supplies the other equality. Division therefore yields

$$P_j = \frac{F_j - \ell_{F_j(z)}}{U_z} \in \text{span}_H\{u^{1-m}, \dots, u^{m-1}\} \subset C_K(H).$$

Since $b \mapsto \ell_b$ is H -linear and $\ell_1 = 1$, we obtain

$$r_0 + \alpha_j r_1 - P_j = \frac{f - \ell_v - \alpha_j}{U_z} - P_j = \frac{f - F_j}{U_z}.$$

No denominator vanishes on the base-field circle, so the claimed agreements follow. Finally $r_1 = -1/U_z$. For any codeword P in the outer Laurent band, clearing $r_1 - P$ by $u^m(uU_z)$ gives a polynomial of degree at most $K + 2$. It is nonzero at $u = a$, where its value is $-a^{m+1}$. This proves the direction bound. $\square$

An explicit family and finite count. The construction in Appendix B supplies such a family: put $f = R_{U_0}$ and

$$R_U = a_0 u^{-(m+2)} G_U(u^2), \quad F_U = f - R_U.$$

The common leading and constant coefficients give support $F_U \subseteq [-m, m]$, while R_U has $K + 4$ roots on the domain. For $(n, K) = (512, 128)$, the exact counts give

$$N = \left\lceil \binom{255}{66} / 256 \right\rceil, \quad J \geq 18,014,398,492,704,768.$$

On every good sample the certified raw proximity event therefore has probability at least $J/p^4 \approx 2^{-70}$ under a uniform linear batching challenge. The good-sample fraction costs approximately one further bit. This is not a protocol security estimate.

Domains and the source sampler. The transfer theorem applies to arbitrary base-field circle domains. The displayed family transports to a coset $u_0\mu_n$ by replacing u with u/u_0 . Reality and Laurent support are preserved, and its quotient witnesses lie in the rotation-invariant interior band. Stwo's canonical domain is precisely such a coset, with u_0 of order $2n$. The source samples using stereographic coordinates $((1 - t^2)/(1 + t^2), 2t/(1 + t^2))$. For uniform $t \in E$, each admissible circle point has exactly one preimage; the two singular parameters and the omitted point are outside this admissible set. Thus, under independent uniform secure-field draws, the joint sample-and-batching proximity event has probability at least $SJ/(2p^8)$.

Scope and checks. Exact generic-sample fixtures pass over $\mathbb{F}_{127}$ and M31; a separate $\mathbb{F}_{23}$ fixture checks the fixed second claim 1 and distinct labels on additive H -cosets. Section C.6 gives a compact description of the large support family and constructs its fixed table efficiently. An efficient algorithm for finding a favorable claim and its witnesses is still not established. Additional columns, periodicity terms, claim constraints from the AIR, commitment consistency, FRI queries and grinding remain to be analyzed. In particular, neither the proximity probability nor the existence of a nearby codeword establishes an accepted false proof.

C.1 A fixed claim and a stronger joint probability bound

The preceding argument selects a favorable claimed value separately for each sample and discards up to half the admissible samples. Both losses can be avoided when the objective is a joint sample-and-batching bound. The following averaging argument fixes a single claim before either challenge. Its guarantee concerns the total number of successful pairs; it does not promise the same number of labels at every sample.

Theorem C.2 (Fixed-claim list-to-quotient transfer). *Under the hypotheses of the list-to-quotient transfer above, let*

$$A_* = \left\lceil \frac{NS^2}{p^2(S + K(N - 1))} \right\rceil, \quad S = (p^2 - 1)^2.$$

There is one value $v_ \in wH$, independent of the sample and batching challenge, such that the fixed tables $(f, 0)$ with fixed claims $(v_*, 1)$ have at least A_* distinct pairs (a, α) with a admissible and $\alpha \in H$, for which*

$$r_{0,a} + \alpha r_{1,a} \text{ agrees with } C_K(H) \text{ in at least } T \text{ coordinates.}$$

For every admissible sample, the direction has at most $K + 2 < T$ agreements with any codeword. Under independent uniform draws $t, \alpha \in E$, with t mapped to the circle by the stereographic sampler, the probability of the certified raw proximity event is at least

$$\frac{A_*}{p^8} \geq \frac{NS^2}{p^{10}(S + K(N - 1))}. \quad (22)$$

The two singular sampler parameters are counted as failures in this unconditioned bound.

Proof. Let $\mathcal{A}$ be the set of admissible parameters. For $a \in \mathcal{A}$ and $b \in E$, let

$$m_{a,b} = \#\{j : F_j(z(a)) = b\}, \quad \mathcal{I} = \{(a, b) : m_{a,b} > 0\}.$$

Then $\sum_{a,b} m_{a,b} = NS$. Each unordered pair of distinct Laurent polynomials collides at at most K parameters, by the same degree- K root bound used above. Consequently

$$\sum_{a,b} m_{a,b}^2 = NS + 2 \sum_a \sum_b \binom{m_{a,b}}{2} \leq NS + KN(N-1).$$

Cauchy–Schwarz, applied once to all incidences, gives

$$|\mathcal{I}| \geq \frac{(NS)^2}{NS + KN(N-1)} = \frac{NS^2}{S + K(N-1)}. \quad (23)$$

Fix the p^2 representatives wH for the additive cosets of H in E . Partition $\mathcal{I}$ according to the coset containing b . One fixed coset $v_* + H$ contains at least A_* incidences. For each of these incidences choose one witness F_j and put $\alpha = b - v_* \in H$. This is an injective map from the retained incidences to pairs (a, α) , even when several polynomials have the same value at one sample. Both v_* and the tables are fixed throughout this count.

For the sample a , write $U_a = U_{z(a)}$ and use the interpolation operator $\ell_{a,b}$ from the preceding proof. The quotient witness is

$$P_{a,j} = \frac{F_j - \ell_{a,b}}{U_a} \in C_K(H), \quad r_{0,a} + \alpha r_{1,a} - P_{a,j} = \frac{f - F_j}{U_a}.$$

Here $r_{0,a} = (f - \ell_{a,v_*})/U_a$ and $r_{1,a} = -1/U_a$. Thus all retained incidences give the required agreements. The degree- $K + 2$ direction argument applies to every admissible sample, regardless of the chosen coset. Finally, every admissible a has exactly one nonsingular stereographic preimage $t \in E$. The A_* distinct pairs therefore occupy A_* of the p^8 equally likely pairs (t, α) , proving the probability claim. $\square$

The same finite instance with fixed claims. For the preceding $(n, K, T) = (512, 128, 132)$ instance,

$$N = \left\lceil \binom{255}{66} / 256 \right\rceil, \quad S = (p^2 - 1)^2, \quad p = 2^{31} - 1.$$

Exact integer arithmetic gives

$$A_* = 766247768292072891947921916124690454684786490150411277, \quad 2^{69} A_* > p^8.$$

The certified joint raw proximity probability is therefore strictly greater than 2^{-69} . The earlier half-sample bound is $SJ/(2p^8)$, approximately 2^{-71} at these same parameters; the improvement is approximately a factor of four. The earlier 2^{-70} figure was conditional on a good sample, so it should not be compared directly with this joint probability.

Quantifiers and limits. This strengthens the timing statement to *there exists one fixed claim, followed by random sample and batching draws*. It does not identify an efficiently computable v_* . Section C.6 constructs the table and describes the support bank implicitly; efficient witness recovery remains open. Individual samples may have fewer labels than the previous per-good-sample bound, or no certified label. The false claim 1 for the zero table remains part of the construction. The result concerns raw quotient agreement under the stated independent uniform challenge model. It does not establish a Fiat–Shamir attack, an accepted false proof, or a 69-bit security level for Stwo.

Reproducible checks. The companion script `checks/verify_fixed_claim.py` checks the displayed large-field counts using exact integers and rational numbers. An independent toy instance over $\mathbb{F}_{7^4}$ exhausts all 2,304 admissible samples. It checks the sampler bijection, counts distinct evaluation incidences and additive cosets, fixes one claim globally, and verifies every retained quotient identity and agreement count. A separate small family with actual evaluation collisions tests the multiplicity term in (23). These finite checks support the formulas; the proof above establishes the general statement.

C.2 Subtracting roots forced by shared agreements

The degree bound in Theorem C.2 counts all roots of a difference as possible sample collisions. Roots on the base-field evaluation domain cannot be admissible samples. Agreements with the same received word force many such roots and give a sharper bound.

Proposition C.3 (Agreement-aware collision bound). *Suppose that, for each of the N polynomials in the transfer theorem, we select T agreement coordinates from a common pool of n_0 base-field circle points. Write*

$$NT = n_0q + r, \quad 0 \leq r < n_0, \quad R = (n_0 - r) \binom{q}{2} + r \binom{q+1}{2}, \quad C_* = K \binom{N}{2} - R.$$

Then the total number of unordered polynomial-pair collisions over admissible samples is at most C_ . In particular, the assumed family implies $C_* \geq 0$. The total number I of distinct sample/value incidences satisfies*

$$I \geq \frac{N^2 S^2}{NS + 2C_*}. \quad (24)$$

Consequently Theorem C.2 holds with the stronger pair count

$$A_{\text{agr}} = \left\lceil \frac{N^2 S^2}{p^2 (NS + 2C_*)} \right\rceil.$$

Proof. Let d_x count selected agreement sets containing coordinate x . Then $\sum_x d_x = NT$, and double counting gives

$$\sum_{j < k} |A_j \cap A_k| = \sum_x \binom{d_x}{2} \geq R.$$

The last inequality follows by balancing the integer counts: moving one unit from a count at least two larger than another strictly decreases the sum of binomial coefficients. At a coordinate of $A_j \cap A_k$, both polynomials equal the received word. These are distinct roots of $u^m(F_j - F_k)$, in addition to any admissible collision roots, and are disjoint from the admissible sample set. Hence pair (j, k) has at most $K - |A_j \cap A_k|$ admissible collisions. Summing proves the bound. The same incidence Cauchy–Schwarz argument as before yields (24); partition into the p^2 additive H -cosets gives the final statement. $\square$

In the displayed two-fiber family, all 132 agreement coordinates avoid the two points above $1 \in \mu_{256}$. Thus $n_0 = 510$ is valid, with $N = \lceil \binom{255}{66} / 256 \rceil$. The exact bound becomes

$$A_{\text{agr}} = 1045232662865816582797566371469227027334159065108360767.$$

The negative base-two logarithm of A_{agr}/p^8 is approximately 68.5521. This is a further factor of approximately 1.3641 over Theorem C.2 at the same parameters.

C.3 Fixed claims under nonuniform batching schedules

The coefficient multiplying the second quotient need not itself be a uniform field element. A second averaging argument handles a prescribed function of the original uniform challenge. Here the claimed value is allowed to range over all of E , rather than just the representatives wH .

Theorem C.4 (Transfer for a prescribed scalar schedule). *Use the same list, admissible samples, and raw quotients as above. Let $g_0, g_1 : E \rightarrow E$ be fixed functions, and set*

$$\mathcal{B} = \{\alpha \in E : g_0(\alpha) \neq 0, g_1(\alpha)/g_0(\alpha) \in H\}, \quad G = |\mathcal{B}|.$$

Let I_0 be any real lower bound for the number of distinct sample/value incidences, for example the right-hand side of (24). There is a single claim $v \in E$, fixed before both challenges, for which at least

$$A_{\text{sched}} = \left\lceil \frac{GI_0}{p^4} \right\rceil$$

distinct admissible pairs (a, α) satisfy

$$g_0(\alpha)r_{0,a} + g_1(\alpha)r_{1,a} \text{ agrees with } C_K(E) \text{ in at least } T \text{ coordinates,}$$

with fixed tables $(f, 0)$ and fixed claims $(v, 1)$. Under the same independent uniform challenge model, the joint probability is at least A_{sched}/p^8 .

Proof. For every distinct incidence (a, b) and every $\alpha \in \mathcal{B}$, assign the triple to

$$v = b - \phi(\alpha), \quad \phi(\alpha) = g_1(\alpha)/g_0(\alpha) \in H.$$

There are GI triples and p^4 possible values of v , so one fixed value receives at least A_{sched} triples. For this fixed v , the map to (a, α) is injective: these two coordinates uniquely determine $b = v + \phi(\alpha)$. Choose a polynomial witnessing each retained incidence. The raw quotient identity gives

$$g_0(\alpha)r_{0,a} + g_1(\alpha)r_{1,a} - g_0(\alpha)P_{a,j} = g_0(\alpha)\frac{f - F_j}{U_a}.$$

The witness $g_0(\alpha)P_{a,j}$ lies in the scalar-extended code $C_K(E)$. Nonzero multiplication preserves agreement coordinates. Counting the distinct sampler preimages proves the probability bound. No injectivity or uniform-output assumption about ϕ is used. $\square$

Additional terms that are codewords. The conclusion is preserved after adding any $B_{a,\alpha} \in C_K(E)$ to the batched word: add the same codeword to its witness. Thus other columns need not have zero quotients if their complete contribution is proved to lie in this code. A common preceding factor α^e , $e > 0$, removes at most the challenge $\alpha = 0$ from the certified set. These statements require code membership of the additional contribution; they do not discard arbitrary additional quotient or AIR terms.

Polynomial schedules over the subfield. If $g_0, g_1 \in H[X]$ and g_0 is a nonzero polynomial of degree $d_0 < p^2$, then $G \geq p^2 - d_0$: every $\alpha \in H$ outside the roots of g_0 qualifies. Thus even a nonlinear relative coefficient incurs no additional loss from the degree of g_1 . More precisely, for $g_0 = 1$, $g_1 = X^d$, $d \geq 1$, cyclicity of E^* gives

$$G = 1 + (p^2 - 1) \gcd(d, p^2 + 1). \quad (25)$$

Indeed a nonzero α qualifies exactly when $\alpha^{d(p^2-1)} = 1$. The assertion is about the number of original challenges, not a claim that α^d is uniformly distributed.

A finite dependence on the gap between powers. For $p = 2^{31} - 1$, each of $d = 1, 2, 5, 10$ divides $p^2 + 1$. The same list with the agreement-aware incidence bound gives:

Power gap d	One-sample columns	Qualifying challenges G	$-\log_2(A_{\text{sched}}/p^8)$
1	2	p^2	68.5521
2	3	$2p^2 - 1$	67.5521
5	6	$5p^2 - 4$	66.2301
10	11	$10p^2 - 9$	65.2301

The final column is approximate; exact integer certificates prove, respectively, probabilities greater than 2^{-69} , 2^{-68} , 2^{-67} , and 2^{-66} . All rows use the same $(n, K, T) = (512, 128, 132)$; the favorable fixed claim may differ by row. At the quotient layer the schedule 1, α^d is realized by putting $d - 1$ zero columns with truthful zero claims between the two designated columns, with one sample per column at the same point. Every inserted numerator is identically zero, while its sample consumes one challenge power. This establishes the displayed quotient expression and column count; it does not show that any particular AIR allows this arrangement.

Corollary C.5 (The periodicity coefficient $\alpha + \alpha^3$). *For $g_0 = 1$ and $g_1 = X + X^3$, one has $G = 2p^2 - 1$. Using the agreement-aware bound at $(n, K, T) = (512, 128, 132)$ gives*

$$A_{\text{cubic}} = \left\lceil \frac{(2p^2 - 1)N^2S^2}{p^4(NS + 2C_*)} \right\rceil, \quad 2^{68}A_{\text{cubic}} > p^8.$$

Its joint raw-proximity lower bound is therefore strictly greater than 2^{-68} ; its negative base-two logarithm is approximately 67.5521.

Proof. Put $q = p^2$ and write $\alpha = x + wy$, with $x, y \in H$ and $w^2 = \kappa$. For $y = 0$ there are q qualifying challenges. For $y \neq 0$, equality of $\alpha + \alpha^3$ with its conjugate is equivalent to

$$3x^2 + \kappa y^2 = -1.$$

The element $3 \in \mathbb{F}_p^*$ is a square in H (here $p \neq 3$). Choose $s \in H$ with $s^2 = 3$. Since $i^2 = -1$,

$$\text{Norm}_{E/H}(sx + wiy) = 3x^2 + \kappa y^2.$$

The nonzero norm fiber at -1 has $q + 1$ elements. Exactly two have $y = 0$, since $-1/3$ is a square in H . Thus $q - 1$ further challenges qualify, giving $G = 2q - 1$. The numerical inequality is certified with exact integers by the companion script. $\square$

The source schedule that gives this coefficient. At the pinned proving revision [15], `build_samples_with_randomness_and_periodicity` assigns powers across trees, columns, and samples in that order. Two consecutive one-sample columns at the same point receive 1, α if they are the first nonempty columns. No periodicity sample is added to either.

There is also an exact three-power realization of the cubic coefficient. Place the (f, v) column first with its sole sample at z . Place the zero column next, with claims 0 at a previous point z_- and 1 at z . Give the zero column the maximal lifted log size. Its periodicity point then coincides with z , and the source assigns the three powers $\alpha, \alpha^2, \alpha^3$ to its periodicity, previous, and current samples. Its previous-sample numerator is zero. The normalized sum is therefore exactly

$$r_{0,z} + (\alpha + \alpha^3)r_{1,z}.$$

If the previous point makes the source’s conjugate-line formula singular, that sample must additionally be excluded. For a fixed base-field rotation $z_- = hz$, at most $p^2 - 1$ initially admissible samples are lost: the excluded parameters satisfy $a\bar{a} = -h^{-2}$, a norm fiber of size $p^2 + 1$, of which two parameters lie in H . Replacing S by the conservative $S' = S - (p^2 - 1)$ in the incidence bound gives a version covering both source denominators. The strict 2^{-68} comparison still holds.

This realizes the scalar schedule at the quotient layer for the stated column arrangement. It does not establish that an AIR admits these claims, that a complete transcript accepts, or that the favorable claim and corresponding witnesses can be found efficiently.

C.4 The radius of the raw-quotient example

The (512, 128, 132) example has rate $\rho = 1/4$, gap $\eta = 1/128$, and radius $\theta = 95/128$. It is *above* the characteristic-based Elias radius: $H_p(95/128) \approx 0.7687496752 > 3/4$. This comparison follows without floating-point assumptions from

$$p^{96}95^{95}33^{33} < (p - 1)^{95}128^{128}.$$

It follows that this raw-quotient example is not itself a counterexample within the below-Elias range of the whitepaper’s list prescription. The below-Elias counterexamples elsewhere in the paper remain separate constructions. Neither the 2^{-69} bound nor its stronger variants should be interpreted as a protocol security level.

C.5 What a nearby codeword supplies at the FRI query stage

The source permits one further conditional statement. Its first FRI layer combines the quotient values computed at sampled positions with complementary values supplied by the FRI proof. Agreement is required at the sampled positions; the surrounding folding block need not consist entirely of agreement coordinates.

Lemma C.6 (Conditional query survival). *Let $R \in E^D$ be the quotient word computed from fixed committed tables and claimed openings, and suppose $P \in C_K(E)$ agrees with R on $A \subseteq D$. Assume that an honest FRI proof is formed for P on this same domain and degree bound, and that the original table decommitments are valid. If every sampled quotient-query position lies in A , the first-layer value substitution leaves the honest FRI verification unchanged. This holds for the source’s supported folding steps and leaf packing.*

Under q_{qry} independent uniform draws from D , with replacement before sorting and deduplication, the probability of this sufficient event is exactly

$$\left(\frac{|A|}{|D|}\right)^{q_{\text{qry}}}.$$

If the preceding raw-proximity event has probability at least π and supplies $|A| \geq T$ on a length- n domain, then this particular two-stage sufficient event has probability at least $\pi(T/n)^{q_{\text{qry}}}$, under independent future query draws.

Proof. The source function `compute_decommitment_positions_and_rebuild_evals` partitions the sorted query positions into folding blocks. At an actual query position it uses the provided quotient value; at any other position in that block it takes a value from the FRI witness. Form the latter values and all authentication paths using the honest proof for P . When every queried position is in A , the supplied quotient value is also P at that position. Thus every reconstructed block, including

any packed leaf, equals its honest counterpart. Its Merkle check, subsequent fold, and all later FRI checks are unchanged.

The raw query draws all land in A with the displayed probability. Sorting and removing duplicates does not change that event. The final statement follows by conditioning on the earlier sample and batching challenges, including the selected witness, and then averaging. $\square$

Numerical consequence and scope. For the (512, 128, 132) example, $T/n = 33/128$. At 70 raw query draws the additional sufficient-event factor is

$$(33/128)^{70} \approx 2^{-136.8924}.$$

For illustration, multiplying this by the certified raw bound for the eleven-column, power-gap-ten schedule gives a two-stage lower bound of approximately $2^{-202.1225}$. This uses 70 as a stated query count; it does not identify this rate-1/4 example with a production parameter set. These are lower bounds for a particular sufficient event, not upper bounds on all possible strategies or proof-system soundness guarantees.

The lemma is conditional on constructing the codeword and honest FRI proof, providing valid table decommitments, and meeting the prescribed domain and degree metadata. It does not establish the AIR conditions, efficient recovery of witnesses or the favorable fixed claim, or a successful native verifier execution. Proof-of-work search and Fiat–Shamir retry costs are not included in this probability. In particular, an inverse one-trial probability is not automatically an attack work factor: reusing a successful prefix and searching later challenges requires its own analysis.

The companion `checks/verify_query_survival.py` enumerates all $8^3 = 512$ ordered query triples for an eight-coordinate word with five agreements. For each of folding steps 1, 2, 3, precisely 125 triples give unchanged reconstructed blocks, including cases where unqueried positions in the block are disagreement coordinates. This checks the reconstruction argument against an explicit vector commitment model; it does not replace a native Merkle or FRI test.

C.6 An explicit table with an exponentially large implicit list

The support family need not be enumerated to construct its received word. On μ_M , equality of the constant coefficients of the root polynomials is a subset-sum condition modulo M . This yields an efficient exact construction of the fixed table used above.

Proposition C.7 (Constructing the table without enumerating its list). *Let λ generate μ_M and consider the h -element subsets of $\{\lambda, \lambda^2, \dots, \lambda^{M-1}\}$. The sizes of all classes with equal constant coefficient, a largest class, and one support in that class can be computed with $O(M^2h)$ additions of integers of $O(M)$ bits, storing $O(Mh)$ such integers and support masks. For the displayed $M = 256$, $h = 66$ instance, the largest count is*

$$N_{\text{exact}} = 471959451942648482921857753174390587612228702156784439968875.$$

The residue 1 attains this count. It gives an explicitly computed 512-entry $M31$ -valued table on a canonical coset and an implicit family of N_{exact} distinct real Laurent polynomials with support in $[-64, 64]$, each agreeing with that table in exactly 132 coordinates.

Proof. Let $D_{j,k,r}$ be the number of k -subsets of $\{1, \dots, j\}$ whose sum is r modulo M . The initial condition is $D_{0,0,0} = 1$, with all other counts zero, and

$$D_{j,k,r} = D_{j-1,k,r} + D_{j-1,k-1,r-j \bmod M}.$$

Updating k in decreasing order permits an in-place computation using $O(Mh)$ counts. Whenever a count first becomes positive, retain one witness support mask from the responsible predecessor, adding j if the second term supplied it. These masks have at most M bits. Every count is at most 2^{M-1} . The stated operation and storage bounds follow. Also $\sum_r D_{M-1,h,r} = \binom{M-1}{h}$, which checks the total count.

The constant coefficient of $G_U(Y) = \prod_{j \in U} (Y - \lambda^j)$ is $(-1)^h \lambda^{\sum_{j \in U} j}$. Thus the residue classes counted by the recurrence are exactly the constant-coefficient classes. Choose a largest class and its retained support U_0 . The earlier construction sets

$$f(u) = a_0 u^{-(m+2)} G_{U_0}(u^2), \quad F_U(u) = a_0 u^{-(m+2)} (G_{U_0}(u^2) - G_U(u^2)).$$

Here $m = 64$, and a_0 is the explicit phase factor from Appendix B. The common leading and constant coefficients put every F_U in $[-64, 64]$. Its difference from f vanishes at precisely the two-point fibers of U , giving 132 agreements. Distinct supports give distinct polynomials. Rotation to the canonical coset preserves these statements. The table requires only G_{U_0} , not the other polynomials in the family. $\square$

A closed form and a fixed defining support. In this instance, every odd residue has exactly

$$N_{\text{exact}} = \frac{\binom{255}{66} + \binom{127}{33}}{256}$$

supports. This also follows from the roots-of-unity filter. For a character of order $d = 2^s > 1$, the support generating function is $(1-t^d)^{256/d}/(1+t)$; its degree-66 coefficient is $A_d = (-1)^{\lfloor 66/d \rfloor} \binom{256/d-1}{\lfloor 66/d \rfloor}$. Thus the count at residue r is

$$\frac{1}{256} \left(\binom{255}{66} + \sum_{d=2,4,\dots,256} c_d(r) A_d \right),$$

where $c_d(r)$ is $d/2$ if $d \mid r$, $-d/2$ if $d/2 \mid r$ but $d \nmid r$, and zero otherwise. For odd r only $d = 2$ contributes, giving the displayed closed form. For even r the $d = 2$ contribution has the opposite sign; the exact inequality $\sum_{d \geq 4} (d/2) |A_d| < 2 \binom{127}{33}$ shows its remaining terms cannot reach the odd-residue count. The script independently checks this character formula against all 256 dynamic-programming counts.

One may fix the defining support outright as $U_0 = \{2, 3, \dots, 68\} \setminus \{40\}$: it has 66 elements and sum $2305 \equiv 1 \pmod{256}$. Hence even the search for a defining support is unnecessary for this particular table.

Concrete artifact. The companion `checks/construct_m31_bank.py` performs 3,759,360 integer updates; the recorded run took approximately 0.6 seconds for this counting stage. It writes all 256 exact residue counts, a defining support, the canonical domain parameters, and the 512 base-field table entries to `checks/explicit_m31_bank.json`. A checked generic-sample fixture also supplies a normalized quotient witness in the interior Laurent band $[-63, 63]$ and verifies all 512 coordinate identities, with exactly 132 agreements.

The fixture's claimed value is chosen to realize one specified sample/challenge pair. It is not asserted to be the favorable fixed claim promised by Theorem C.4. The family is now explicit as a compact membership rule and exact count, and its fixed table is fully constructed. Finding the favorable claim and recovering a witness efficiently for a fresh challenge remain open. The numerical probability bounds above may continue to use the slightly smaller $N = \lceil \binom{255}{66} / 256 \rceil$, by retaining that many members of the class.

C.7 Witness recovery as a constrained subset product

The implicit description of the family permits an exact search for a witness after sampling. The following reduction isolates the remaining algorithmic problem and gives a complete, although expensive, baseline. The claim v is held fixed throughout the search.

Proposition C.8 (Exact recovery criterion and a complete search). *Use the two-fiber family of Section C.6, with $h = m + 2$, defining support U_0 , residue r , and relative sample parameter $a \notin H$. Assume $\mu_{2M} \subset H$. For any desired value $\beta \in E$, a support U in this family satisfies $F_U(a) = \beta$ if and only if*

$$|U| = h, \quad \sum_{j \in U} j \equiv r \pmod{M}, \quad \prod_{j \in U} (a^2 - \lambda^j) = G_{U_0}(a^2) - \frac{a^h}{a_0} \beta. \quad (26)$$

Each factor on the left is nonzero. A zero target therefore has no solution. For a nonzero target, partition $\{1, \dots, M-1\}$ into parts of sizes s and $M-1-s$. There is a complete search using a table of at most

$$R = \sum_{k=0}^{\min(h, M-1-s)} \binom{M-1-s}{k}$$

right-hand subset records and at most

$$L = \sum_{k=0}^{\min(h, s)} \binom{s}{k}$$

dictionary lookups per target, in addition to constructing the subset products. Keeping all records with equal keys recovers every matching support; keeping one per key suffices to find a witness when one exists.

Proof. Substitute $F_U(a) = a_0 a^{-h} (G_{U_0}(a^2) - G_U(a^2))$ and rearrange. If $a^2 = \lambda^j$, then a is one of its two square roots in $\mu_{2M} \subset H$, contradicting $a \notin H$.

For a right-hand subset V , store its support under the key

$$\left(|V|, \sum_{j \in V} j \bmod M, \prod_{j \in V} (a^2 - \lambda^j) \right).$$

For each left-hand subset W , with cardinality k , residue t , and nonzero product P_W , look up $(h-k, r-t \bmod M, \tau/P_W)$, where τ is the target in (26). The parts are disjoint, so every returned union has the required cardinality, residue, and product. Conversely, each solution has exactly one such decomposition, which the search visits. The subset counts give the stated bounds. $\square$

For the scheduled transfer, set $\beta = v + g_1(\alpha)/g_0(\alpha)$ when $g_0(\alpha) \neq 0$ and this ratio belongs to H . A recovered support supplies F_U , and exact polynomial division then supplies the interior-band quotient witness from Theorem C.4. Preparing the subset-product table depends on a and must generally be repeated when the sample changes. It can be reused for multiple targets at the same sample. This reduction does not identify the favorable fixed claim from the averaging theorem.

Fresh-query check on a smaller valid circle code. The companion `checks/verify_witness_search.py` fixes $v = 0$ before generating samples. It uses $p = 23$, $n = 24$, $M = 12$, $h = 4$, $m = 2$, and the full base-field circle, with defining support $U_0 = \{1, 2, 3, 7\}$ and residue 1. The resulting family has 27 members of outer Laurent degree at most 2, each with 8 agreements. The tower $H = \mathbb{F}_{23}[i]$, $E = H[w]$, $w^2 = 2 + i$, is a field: the script checks $(2 + i)^{(23^2 - 1)/2} = -1$. This is a circle-code experiment; its length is not a Stwo FFT size.

The deterministic experiment chooses 128 distinct admissible samples using a recorded pseudorandom seed. At each sample it tests every one of the 529 challenges in H , together with eight specified outside- H controls. On all 68,736 queries, the returned support set agrees exactly with an independent enumeration of all 27 family members. There are 135 successful queries: 128 have $\alpha = 0$, which always recovers $F_{U_0} = 0$, and seven have $\alpha \neq 0$. No outside- H control succeeds in this fixture. For all 135 recovered witnesses, exact division yields coefficients in H with Laurent support $[-1, 1]$; all 3,240 coordinate identities pass, with exactly eight agreements per witness. The split uses 31 left-hand and 57 right-hand records per sample. These challenges are deliberately exhaustive in H ; the observed success fraction is not an estimate for uniform challenges in E .

Cost at the displayed M31 parameters. For the direct implementation with $s = 127$, $M = 256$, and $h = 66$, the complete right-hand table would have

$$R = \sum_{k=0}^{66} \binom{128}{k} = 228210150822262702568674774219878072803 \approx 2^{127.4236}$$

records, before accounting for bytes per record. Thus this baseline does not give a practical M31 decoder. This is the cost of the specified complete enumeration, not a lower bound on witness-search complexity or a claim that more selective algorithms cannot improve it. The useful conclusion is an exact algebraic target and a checked recovery procedure for fresh queries with a claim fixed in advance.

D Checks and provenance

D.1 Exact interval certificates

The rounded comparisons in the main text follow from Lemma 2.1 and Proposition 2.3. The following table records the integers needed to check the three list examples. All have $k = n/2$.

p	n	s	Certified strict lower bound on $\log_2 L$
$2^{127} - 1$	64	2	41
$2^{521} - 1$	4096	8	3634
$2^{521} - 1$	23444	45	11567

For the first row the original denominator product is 32,285,041, giving 50,186,958,429 codewords. The conditional dispersion certificate (6) strengthens this to 2,674,048,230,075, yielding the displayed exponent 41. For every row, exact integer arithmetic gives $p^s > 2^n$, certifying the sufficient below-Elias condition $\eta \log_2 p > 1$. For the last row, a bound $2^{c_2/\eta}$ would require

$$c_2 > \frac{11567 \cdot 45}{23444} > 22.2.$$

The finite rational-line example uses the conditional first-moment count

$$N = 8,634,334,451,040,106$$

from Section 2.3. The exact inequality $2^{127} - 1 > 64 + 32\binom{N}{2}$ certifies a pole with all N scalar labels distinct. These integers are retained here for reproducibility; the comparisons in the main text only need their rounded powers of two.

D.2 Certificates at additional rates

For Corollary 6.2, the exact guaranteed list size is

$$L = \left\lceil \binom{63}{17} / 64 \right\rceil = 15,827,733,944,904, \quad \eta = \frac{1,048,575}{33,554,432}.$$

Integer arithmetic verifies $p^{100} > 2^{3099}$, $3^{100000} > 2^{158496}$, and $L^{2000} > 2^{87695}$. Since $H_2(1/4) = 2 - \frac{3}{4} \log_2 3 < 0.81128$, these give

$$\log_2 L - \frac{H_2(1/4)}{\eta} > \frac{87695}{2000} - \frac{81128}{100000\eta} = \frac{937767635977}{52428750000} > 17.8865.$$

This implies the rounded ratio in the corollary. For the Elias check, write $\theta = 3/4 - \eta$. The exact rational inequality

$$\eta \frac{3099}{100} > 1 - 2(\theta - 1/2)^2 \geq H_2(\theta)$$

and $\log_2 p > 3099/100$ imply $H_p(\theta) < 3/4$. The same construction at rates $1/8$, $1/16$, and $3/4$ gives list ratios exceeding 2^{13} , 2^{15} , and 2^{13} , respectively. The rate- $3/4$ space has even dimension but is not a native power-of-two FFT size. No such restriction is needed for the mathematical code.

For Corollary 6.3, put $Q = p^4 - p^2$. The exact counts are

$$N = \left\lceil \frac{\binom{255}{36}}{256p^2} \right\rceil = 73,497,635,817,391,996,020,834,$$

$$J = \left\lceil \frac{NQ}{Q + 32(N-1)} \right\rceil = 73,497,635,817,383,868,141,347.$$

Here $\eta = 655359/33554432$. Integer arithmetic verifies $7^{100000} > 2^{280735}$, $7^{1000} < 2^{2808}$, and $J^{1000} > 2^{75960}$. Set

$$H^+ = 3 - \frac{7 \cdot 280735}{8 \cdot 100000} > H_2(1/8).$$

Concavity gives $H_2(7/8 - \eta) < H^+ + \eta(2808/1000)$, whereas exact rational arithmetic gives $\eta(3099/100 - 2808/1000) > H^+$. Together with $\log_2 p > 3099/100$, this certifies the strict Elias bound. Finally,

$$\log_2 J - 26 - \frac{H_2(1/8)}{\eta} > \frac{75960}{1000} - 26 - \frac{H^+}{\eta} = \frac{9064121863}{409599375} > 22.129.$$

After puncturing, the exact parameters are $\rho' = (2^{23} - 1)/(2^{26} - 2)$ and $\eta' = (5 \cdot 2^{18} - 1)/(2^{26} - 2)$. Exact rational logarithm intervals separately certify the Elias bound and a ratio greater than 2^{22} at these changed parameters.

For the quadratic-field example, put $Q_2 = p^2 - 128$. Then

$$N_2 = \left\lceil \binom{127}{18} / 128 \right\rceil = 25,460,924,804,916,891,029,$$

$$J_2 = \left\lceil \frac{N_2 Q_2}{Q_2 + 16(N_2 - 1)} \right\rceil = 285,003,988,493,147,037.$$

Here $\eta = 786431/33554432$. The same H^+ above satisfies $\eta(3099/100 - 2808/1000) > H^+$, proving the strict Elias bound. The integer inequality $J_2^{10000} > 2^{579837}$ gives

$$\log_2 J_2 - 26 - \frac{H_2(1/8)}{\eta} > \frac{579837}{10000} - 26 - \frac{H^+}{\eta} = \frac{13827665031}{1572862000} > 8.791.$$

Thus the ratio exceeds 400. An independent fixture over $\mathbb{F}_{312}$ checks all 945 allowable poles for an anchored coefficient class of 87 supports. It verifies the collision bound and all witness identities for a best pole and for the allowed pole zero; these yield 87 and 84 distinct labels, respectively.

D.3 Certificates for the compact examples

Corollary 6.4 uses the same integers J_2 and J as the two eighth-rate constructions above. The two new Elias comparisons follow from

$$\begin{aligned} p^{896} 873^{873} 151^{151} &> (p-1)^{873} 1024^{1024}, \\ p^{3584} 3505^{3505} 591^{591} &> (p-1)^{3505} 4096^{4096}. \end{aligned}$$

Taking logarithms gives $H_p(873/1024) < 7/8$ and $H_p(3505/4096) < 7/8$. Also,

$$J_2^2 > 2^{115}, \quad J^3 > 2^{227}, \quad 7^5 > 2^{14}.$$

The last inequality gives $H_2(1/8) < 11/20$, so the logarithms of the two ratios exceed

$$\frac{115}{2} - 11 - \frac{11}{20} \frac{1024}{23} > 22, \quad \frac{227}{3} - 13 - \frac{11}{20} \frac{4096}{79} > 34.$$

All these comparisons use exact integer or rational arithmetic.

D.4 Verification and source notes

The accompanying files give exact integer certificates for the finite bounds and exhaustive small-field checks of the constructions. They also include a small explicit test of the power-fiber lift. The proofs establish the asymptotic claims; the experiments check formulas and boundary cases. Large circle lists and suitable poles are obtained by counting, rather than enumerated.

The numerical certificates and construction checks are included with this draft. They use exact integer or finite-field arithmetic. The protocol-related models do not constitute native verifier executions. The source comparisons concern the cited repository revisions; they are not claims about later implementations. This is a working draft with machine-assisted internal checks, and it has not undergone external peer review.

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